



ក្រសួងអប់រំ យុវជន និងកីឡា
វិទ្យាស្ថានបច្ចេកវិទ្យាភ្នំពេញ



ដេប៉ាតឺម៉ង់គណិតវិទ្យាអនុវត្តន៍ និងស្ថិតិ

គម្រោងសញ្ញាប័ត្រវិស្វកម្ម

ប្រធានបទ : ប្រព័ន្ធថាវកម្មសន្តិសុខសង្គមស្របច្បាប់ប្រើប្រាស់ប្រចាំថ្ងៃ
នៅកម្ពុជា

និស្សិត : តាំង គីងម៉េង

ឯកទេស : វិស្វកម្មគណិតវិទ្យាអនុវត្តន៍ និងស្ថិតិ (វិទ្យាសាស្ត្រ
ទិន្នន័យ)

គ្រូទទួលបន្ទុក : លោក ឈាង វីញ៉ា

ឆ្នាំសិក្សា : ២០២៥-២០២៦

**MINISTÈRE DE L'ÉDUCATION,
DE LA JEUNESSE ET DES SPORTS**

INSTITUT DE TECHNOLOGIE DU CAMBODGE

**DÉPARTEMENT DE MATHÉMATIQUES APPLIQUÉES ET
STATISTIQUES**

MÉMOIRE DE FIN D'ÉTUDES D'INGÉNIEUR

Titre : Système de Calcul Quotidien de l'Indice des Prix à la
Consommation (IPC) au Cambodge

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Spécialité : Ingénierie en Mathématiques Appliquées et Statistiques (Data
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Année : 2025-2026



ក្រសួងអប់រំ យុវជន និងកីឡា
វិទ្យាស្ថានបច្ចេកវិទ្យាភ្នំពេញ



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គម្រោងសញ្ញាប័ត្រវិស្វកម្ម

រមស័និស្សិតៈ តាំង ឆ័ងម៉េង

កាលបរិច្ឆេទការពារនិក្ខេបបទៈ ថ្ងៃទី ៣០ ខែ តុលា ឆ្នាំ ២០២៦

អនុញ្ញាតឱ្យការពារគម្រោង

នាយកវិទ្យាស្ថានៈ _____

រាជធានីភ្នំពេញ, ថ្ងៃទី ៣០ ខែ តុលា ឆ្នាំ ២០២៦

ប្រធានបទ : ប្រព័ន្ធថាវិកសម្រាប់ស្ថាប័នបង្រៀនប្រើប្រាស់
ប្រចាំថ្ងៃនៅកម្ពុជា

សហគ្រាស : អគ្គនាយកដ្ឋានសេដ្ឋកិច្ចឌីជីថល (GDDE/MEF)

ប្រធានដេប៉ាតឺម៉ង់ : បណ្ឌិត ភោគ សុខឌី

សាស្ត្រាចារ្យណែនាំ : លោក ឈាង វីញ៉ា

អ្នកទទួលខុសត្រូវ : លោក ឈាង វីញ៉ា

រាជធានីភ្នំពេញ, ២០២៦



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**INSTITUT DE TECHNOLOGIE DU CAMBODGE
DÉPARTEMENT DE MATHÉMATIQUES APPLIQUÉES ET STATISTIQUES**

**MÉMOIRE DE FIN D'ÉTUDES D'INGÉNIEUR
PRÉSENTÉ PAR *M. TAING KORNGMENG***

Date de soutenance : le 30 Octobre 2026

« Autorise la soutenance du mémoire »

Directeur de l'Institut : _____

Phnom Penh, le 30 Octobre 2026

Titre : Système Quotidien de Calcul de l'IPC au Cambodge

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PHNOM PENH, 2026

GRADUATION THESIS REPORT ON
**“CAMBODIA DAILY CONSUMER PRICE INDEX (CPI) PIPELINE:
HIGH-FREQUENCY WEB SCRAPING, SEMANTIC ENTITY MATCHING,
COICOP CLASSIFICATION, AND MACHINE LEARNING FORECASTING”**

at

**General Department of Digital Economy (GDDE), Ministry of Economy and
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Defense Date: 30 September 2026

In partial fulfillment of the requirements for the Degree of
**Engineering in Applied Mathematics and Statistics
(Data Science)**



**Department of Applied Mathematics and Statistics
Institute of Technology of Cambodia**

ACKNOWLEDGEMENT

First of all, I would like to express my deepest gratitude to esteemed and honorable **H.E. Prof. Dr. PO Kimtho**, Director General of the Institute of Technology of Cambodia (ITC), for his visionary leadership, continuous support, and dedication to fostering an advanced academic and research environment in data science and engineering.

I am profoundly grateful to **Dr. LIN Mongkolser**y, Director of the Graduate School, and **Dr. PHAUK Sokkhey**, Head of the Department of Applied Mathematics and Statistics (AMS), for their academic leadership, high standards of excellence, and continuous encouragement throughout my academic trajectory.

I extend my sincere appreciation and special gratitude to my academic advisor, **Mr. CHHEANG Vinha**, for his exceptional mentorship, technical guidance, and valuable econometric insights throughout the conceptualization, system architecture design, and machine learning implementation of this thesis project. His constructive feedback and high engineering standards were instrumental in bringing this work to fruition.

I am profoundly grateful to my internship supervisor, **Mr. LUK Chamnan**, and co-supervisor, **Mr. SEK Sovannarath**, at the General Department of Digital Economy (GDDE), for their dedicated supervision, professional guidance, and practical support throughout my internship within the Ministry of Economy and Finance (MEF). Their valuable domain expertise in digital economic platforms, high-frequency public data systems, and national policy analytics provided an exceptionally enriching operational environment for this research.

Finally, I express my wholehearted appreciation to my parents, family members, and peers in the Department of Applied Mathematics and Statistics. Their unconditional support, patience, and motivation were the ultimate pillars of strength that enabled me to complete this engineering milestone.

អត្ថបទសង្ខេប

គម្រោងស្រាវជ្រាវនេះបង្ហាញពីការរចនា ការអភិវឌ្ឍ និងការដាក់ឱ្យដំណើរការនូវប្រព័ន្ធវាស់វែងសន្ទស្សន៍ថ្លៃទំនិញប្រើប្រាស់ប្រចាំថ្ងៃនៅកម្ពុជា (Cambodia Daily CPI Medallion Pipeline) ដោយប្រើប្រាស់បច្ចេកវិទ្យាប្រមូលទិន្នន័យពីគេហទំព័រ (High-Frequency Web Scraping), ដំណើរការភាសាធម្មជាតិ (NLP), បញ្ញាសិប្បនិម្មិត (AI) និងវិធីសាស្ត្រស្ថិតិសេដ្ឋកិច្ចវិមាត្រ ដើម្បីដោះស្រាយបញ្ហាគម្លាតពេលវេលានៃការចេញផ្សាយរបាយការណ៍អតិផរណាប្រចាំខែបែបប្រពៃណី។

ប្រព័ន្ធនេះធ្វើការប្រមូលទិន្នន័យថ្លៃទំនិញប្រចាំថ្ងៃដោយស្វ័យប្រវត្តពីប្រភពទីផ្សារអនឡាញធំៗចំនួន ២០ ប្រភពក្នុងរាជធានីភ្នំពេញ គ្របដណ្តប់លើទំហំ ១២ ជំពូកនៃចំណាត់ថ្នាក់ការប្រើប្រាស់តាមគោលបំណងរបស់បុគ្គល (UN COICOP) តាមរយៈ Apache Airflow និងរក្សាទុកក្នុងស្ថាបត្យកម្មទិន្នន័យ Medallion Architecture (Bronze, Silver, Gold) លើប្រព័ន្ធគ្រប់គ្រងមូលដ្ឋានទិន្នន័យ PostgreSQL 16។ នៅក្នុងស្រទាប់ Silver ដំណើរការសម្អាតទិន្នន័យ និងស្តង់ដារភាសាត្រូវបានអនុវត្ត រួមផ្សំជាមួយម៉ូដែល Dense Vector Embeddings (text-embedding-004), Deterministic Specification Guards និង Large Language Models (Google Gemini) ដើម្បីធ្វើការផ្គុំផ្គង់អត្តសញ្ញាណទំនិញស្វ័យប្រវត្ត (Entity Resolution) និងចាត់ថ្នាក់ទំនិញចូលជំពូក COICOP ទំហំ ១២ តាមរយៈយន្តការ ៤ កម្រិត (4-Tier Classification Ladder)។

នៅក្នុងស្រទាប់ Gold ការគណនាសន្ទស្សន៍ត្រូវបានអនុវត្តស្របតាមស្តង់ដារអន្តរជាតិរបស់ IMF/ILO ដោយប្រើរូបមន្តមធ្យមធរណីមាត្រ Jevons Geometric Mean នៅកម្រិតទំនិញមូលដ្ឋាន ដើម្បីលុបបំបាត់លម្អៀងរូបមន្ត (Formula Bias) រួមជាមួយយន្តការបំពេញទិន្នន័យបាត់បង់ ៧ ថ្ងៃ (7-Day Carry-Forward Imputation) និងការបូកសរុបកម្រិតម៉ាក្រូតាមរូបមន្ត Modified Laspeyres ដោយផ្អែកលើទម្ងន់ចំណាយជាតិរបស់វិទ្យាស្ថានជាតិស្ថិតិ (NIS) សម្រាប់ទំហំ Headline CPI និង Core CPI។ លើសពីនេះ គំរូទស្សន៍ទាយម៉ាស៊ីនរៀន (Prophet, XGBoost, LSTM) ត្រូវបានអភិវឌ្ឍជាមួយអថេរអត្រាប្តូរប្រាក់ និងតម្លៃប្រេងឥន្ធនៈ ដើម្បីផ្តល់នូវការព្យាករណ៍អតិផរណារយៈពេលខ្លីប្រកបដោយភាពត្រឹមត្រូវខ្ពស់ និងបង្ហាញលទ្ធផលតាមរយៈ Metabase និង Power BI Dashboards សម្រាប់គាំទ្រការរៀបចំគោលនយោបាយម៉ាក្រូសេដ្ឋកិច្ច។

ពាក្យគន្លឹះ៖ សន្ទស្សន៍ថ្លៃទំនិញប្រើប្រាស់ (CPI), ការប្រមូលទិន្នន័យស្វ័យប្រវត្ត (Web Scraping), ស្ថាបត្យកម្ម Medallion, ចំណាត់ថ្នាក់ COICOP, បញ្ញាសិប្បនិម្មិត, រូបមន្ត Jevons, ការព្យាករណ៍អតិផរណា។

ABSTRACT

The Consumer Price Index (CPI) serves as the primary macroeconomic indicator for measuring changes in the cost of living, calibrating monetary policy, and adjusting social safety nets. However, traditional manual monthly survey methodologies suffer from substantial temporal publication lags (30 to 60 days), intra-month volatility blind spots, and limited sampling granularity. In an economy characterized by high dollarization, strong import dependency, and extreme expenditure concentration in food and fuel ($> 74\%$ of the national basket), delayed macroeconomic intelligence hinders timely policy intervention by institutions such as the Ministry of Economy and Finance (MEF) and the National Bank of Cambodia (NBC).

To overcome these structural limitations, this thesis presents the design, implementation, and empirical validation of the **Cambodia Daily Consumer Price Index (CPI) Medallion Pipeline**—an automated, production-grade big data and machine learning system for real-time inflation tracking and forecasting. The pipeline orchestrates 20 daily automated web crawlers via Apache Airflow 2.9.3, capturing high-frequency microdata across all 12 United Nations Classification of Individual Consumption According to Purpose (COICOP) divisions in Phnom Penh.

Operating on a Medallion Star-Schema architecture within PostgreSQL 16, raw JSONB records in the Bronze layer undergo text normalization, Khmer numeral transliteration, and metric unit conversion in the Silver layer. Longitudinal entity resolution is achieved via a hybrid engine combining 768-dimensional dense vector embeddings (`text-embedding-004`) with deterministic specification guards, achieving high semantic precision across noisy retail titles. Product classification into the 12 COICOP divisions is governed by a 4-tier ladder combining human overrides, store domain locks, vector centroid matching, and batched Google Gemini LLM inference with PostgreSQL caching.

In the Gold aggregation layer, micro-level daily prices are compiled using the axiomatic unweighted geometric mean (**Jevons formula**) coupled with an international ILO-standard 7-day carry-forward and class-mean imputation engine, completely eliminating arithmetic price bounce bias. Macro-level aggregation utilizes a Modified Laspeyres formulation weighted by official National Institute of Statistics (NIS) expenditure shares to publish daily Headline CPI (100.000% basket) and Refined Core CPI (52.200% basket). Furthermore, the system deploys an intra-month inflation nowcasting engine chain-linked to official NIS benchmarks alongside gradient-boosted decision tree models (LightGBM) with walk-forward cross-validation, achieving substantial reductions in Root Mean Square Error (RMSE) over standard autoregressive benchmarks for multi-horizon forecasting ($H \in \{7, 14, 30\}$ days). Real-time indicators are served via containerized Metabase and Power BI executive dashboards, establishing a replicable blueprint for modern digital public data infrastructure in developing Southeast Asian economies.

Keywords: Consumer Price Index (CPI), High-Frequency Web Scraping, Medallion Architecture, COICOP Classification, Large Language Models, Jevons Geometric Mean, Inflation Nowcasting, Machine Learning Forecasting, LightGBM.

LIST OF ABBREVIATIONS

AI	Artificial Intelligence	LLM	Large Language Model
AMS	Applied Math & Statistics	LSTM	Long Short-Term Memory
API	Application Prog. Interface	MAE	Mean Absolute Error
ARDL	Autoregressive Dist. Lag	MAPE	Mean Absolute Pct. Error
BERT	Bidirectional Encoder (NLP)	MDI	Mean Decrease in Impurity
BI	Business Intelligence	MEF	Ministry of Economy & Finance
BIS	Bank for Int. Settlements	MIDAS	Mixed-Data Sampling
BPP	Billion Prices Project	ML	Machine Learning
CI/CD	Cont. Integration / Deploy.	MLOps	Machine Learning Operations
COICOP	UN Purpose Classification	MLP	Multilayer Perceptron
CPI	Consumer Price Index	MOC	Ministry of Commerce
CSES	Cambodia Socio-Econ. Survey	MoM	Month-on-Month
DAG	Directed Acyclic Graph	MSFE	Mean Squared Forecast Error
dbt	Data Build Tool	NBC	National Bank of Cambodia
DOM	Document Object Model	NIS	National Institute of Statistics
DoD	Day-on-Day	NLP	Natural Language Processing
ECB	European Central Bank	NSI	National Statistical Institute
EDC	Electricité du Cambodge	OECD	OECD
EDA	Exploratory Data Analysis	OLS	Ordinary Least Squares
ERD	Entity Relationship Diagram	ONS	Office for National Statistics
FX	Foreign Exchange	PFI	Permutation Feature Import.
GBRT	Gradient Boosted Trees	PPWSA	Phnom Penh Water Supply
GDDE	Gen. Dept. Digital Economy	QU-GEKS	Quality-Adjusted GEKS
GEKS	GEKS Multilateral Index	RBAC	Role-Based Access Control
GRU	Gated Recurrent Unit	REST	Representational State Transf.
GTIN	Global Trade Item Number	RMSE	Root Mean Square Error
HITL	Human-in-the-Loop	SCD	Slowly Changing Dimension
HICP	Harmonised Index CPI	SHAP	SHapley Additive exPlanations
ILO	Int. Labour Organization	SKU	Stock Keeping Unit
IMF	International Monetary Fund	SQL	Structured Query Language
IQR	Interquartile Range	SVR	Support Vector Regression
ITC	Inst. of Technology Cambodia	TF-IDF	Term Freq. - Inv. Doc. Freq.
JSON	JavaScript Object Notation	UN	United Nations
JWT	JSON Web Token	USD	United States Dollar
KHR	Cambodian Riel	WBS	Work Breakdown Structure
LightGBM	Light Gradient Boosted Machine	XGBoost	Extreme Gradient Boosting
		YoY	Year-on-Year

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I. INTRODUCTION

1.1 Presentation of Internship

This graduation thesis report formally documents the applied data science research and engineering system developed during the graduation internship program in Data Science within the Department of Applied Mathematics and Statistics (AMS) at the Institute of Technology of Cambodia (ITC). The internship program serves as a critical academic and professional milestone designed to bridge advanced theoretical econometric modeling, modern cloud data engineering architectures, natural language processing, and machine learning with national-scale real-world applications.

1.1.1 Objective of Internship

The overarching objective of the internship is to apply cutting-edge data science methodologies to solve macroeconomic sensing and high-frequency inflation monitoring challenges in Cambodia. The program focuses on converting academic knowledge—encompassing multi-layer data warehousing, dense vector representations, Large Language Models (LLMs), micro-econometric index aggregation, and time-series forecasting—into a resilient, production-grade automated software system capable of serving national policy institutions.

1.1.2 Duration and Administrative Details

The structured internship was executed over a two-month operational timeline from August 03, 2026, to September 30, 2026. The institutional and supervisory details are summarized in Table 1.1.

Attribute	Internship Specification
Start Date	03 August 2026
End Date	30 September 2026
Duration	2 Months (Full-Time)
Internship Position	Data Science Intern
Author / Student	TAING KORNGMENG
Academic Advisor	Mr. CHHEANG Vinha
Company Supervisor	Mr. LUK Chamnan
Company Co-Supervisor	Mr. SEK Sovannarath
Host Organization	General Department of Digital Economy (GDDE)
Parent Ministry	Ministry of Economy and Finance (MEF), Cambodia
Degree Program	Engineering in Applied Mathematics and Statistics (Data Science)
Academic Institution	Institute of Technology of Cambodia (ITC)

Table 1.1: Internship Administrative Framework

1.2 Presentation of Host Organization

The internship was conducted at the **General Department of Digital Economy (GDDE)**, a key government institution operating under the **Ministry of Economy and Finance (MEF)**, Cambodia.

1.2.1 Organization Profile

The GDDE leads Cambodia’s national digital transformation agenda by developing digital policies, managing national data platforms (such as the DataEF portal), and applying data-driven solutions to enhance economic management and public service efficiency, in direct alignment with the *Cambodia Digital Economy and Society Policy Framework 2021–2035*.

1.2.2 Organization History and Strategic Roadmap

Established by a 2023 Royal Sub-Decree, the GDDE was created to centralize and accelerate the implementation of Cambodia’s national digital economy strategy. The strategic development follows a clear three-stage national roadmap:

- **2021–2025 (Foundations):** Building digital foundations, regulatory frameworks, open data ecosystems, and core digital public infrastructure.
- **2026–2030 (Expansion):** Expanding digital transformation and strengthening high-frequency digital economic services across all socio-economic sectors.
- **2031–2035 (Global Competitiveness):** Developing an advanced, highly integrated, and globally competitive digital economy and society.

1.2.3 Vision and Mission

- **Vision:** To be the primary driving force behind Cambodia’s digital transformation, fostering a modern economy where data, technology, and innovation create boundless opportunities for citizens, businesses, and sustainable national growth.
- **Mission:** To lead, manage, and develop Cambodia’s digital economy in an inclusive, equitable, and sustainable manner by leveraging technology, innovation, and data-driven approaches to enhance economic efficiency and public service delivery.

1.2.4 Key Responsibilities and Strategic Initiatives

- **Policy & Regulation:** Formulating technical guidelines, data governance standards, and regulatory frameworks to foster the digital economy and e-commerce ecosystem.
- **National Data Management:** Developing and managing centralized public platforms (such as the **DataEF** open data portal) to provide high-quality economic and financial data to policymakers and researchers.
- **Digital Infrastructure Oversight:** Directing the technical architecture and rollout of major national systems, including the **CamInvoice** electronic invoicing platform and digital public investment management tools.
- **Innovation & Ecosystem Development:** Collaborating with public and private stakeholders to support digital startups, accelerate SME digitalization, and maintain a secure digital commerce environment.

1.3 Macroeconomic Background and Motivation

The Consumer Price Index (CPI) is the fundamental macroeconomic metric used worldwide to quantify changes in the cost of living, track inflationary pressures, guide central bank monetary policy, adjust public wages, calibrate social safety nets, and index national financial contracts. In Cambodia, the official Consumer Price Index is compiled and published by the National Institute of Statistics (NIS) within the Ministry of Planning, adhering to the United Nations Classification of Individual Consumption According to Purpose (COICOP) hierarchy.

1.3.1 Structural Dynamics of the Cambodian Economy

Monitoring consumer price movements in Cambodia is uniquely complex due to specific structural characteristics of the domestic economy:

1. **Extensive Dollarization and Dual-Currency Dynamics:** The Cambodian retail economy operates under high dollarization, with both Cambodian Riel (KHR) and US Dollars (USD) circulating simultaneously. E-commerce platforms, supermarkets, and service providers price goods in both currencies. Consequently, daily exchange rate fluctuations directly and immediately pass through to the retail cost of imported consumer goods.

2. **Extreme Expenditure Concentration in Food and Energy:** According to the official NIS Cambodia Socio-Economic Survey (CSES 2023) expenditure weights, **Division 01 (Food and Non-Alcoholic Beverages) accounts for 44.775%** of the total national consumer basket. Within food, basic staples dominate household expenditures: **Rice and Cereals alone represent 18.50%**, Meat accounts for **10.20%**, and Fresh Fish represents **5.80%**. Furthermore, **Division 04 (Housing, Utilities, Gas) contributes 17.084%** and **Division 07 (Transport and Fuel) contributes 12.180%**. Because over 74.039% of the national basket is concentrated in food, utilities, and transport, global supply chain shocks or regional fuel spikes create acute price surges that disproportionately impact vulnerable households.
3. **Import Dependency for Essential Consumer Goods:** Cambodia relies heavily on imported refined petroleum fuels, packaged pharmaceuticals, telecommunications equipment, and manufactured household goods. Changes in international logistics costs or foreign exchange rates impact domestic shelves with high velocity.

1.4 Problem Statement

The Consumer Price Index (CPI) is essential for monitoring inflation and the cost of living in Cambodia. However, traditional manual price collection is slow, resource-intensive, and released only monthly, meaning rapid intra-month and daily price changes go undetected by policymakers. Meanwhile, many businesses and e-commerce platforms publish prices online, but this data is scattered across platforms and fluctuates daily, making manual tracking impractical.

Although web-scraped retail data offers high frequency and granular visibility, transforming raw web listings into a reliable daily Consumer Price Index poses four core operational challenges:

- Problem 1: Information Lag vs. Daily Volatility:** Official monthly statistical releases have a 30 to 60-day lag, leaving institutions like the MEF and the National Bank of Cambodia (NBC) blind to high-frequency fuel and food price shocks.
- Problem 2: Noisy Free-Text Data and Entity Matching:** E-commerce listings lack uniform barcodes (GTINs) and have fluctuating titles, causing false merges across different pack sizes or storage specs.
- Problem 3: Multilingual COICOP Classification Complexity:** Categorizing thousands of multilingual retail listings (Khmer, English, French) into the 12 UN COICOP divisions is intractable via static manual rules.
- Problem 4: Shrinkflation and Formula Drift:** Changes in package sizes and standard arithmetic formula bias (**Carli index**) introduce artificial drift into long-term inflation tracking.

1.5 Project Goal and Research Objectives

1.5.1 Project Goal

The main goal of this project is to develop an automated system that collects consumer price data from online sources in Cambodia on a daily basis and uses the collected data to support Consumer Price Index (CPI) estimation, price trend monitoring, and machine learning inflation forecasting.

1.5.2 General Objective

The objective of this project is to design and implement an end-to-end data platform for collecting, storing, processing, monitoring, and analyzing price data related to selected COICOP basket categories. The system supports web-based data collection, structured Medallion data storage, query processing, data quality monitoring, visualization, and machine learning models for CPI estimation and inflation analysis.

1.5.3 Specific Technical Objectives

To achieve this goal, the research is organized around four specific operational objectives:

1. **Build an Automated Data Infrastructure and Monitoring System:** Set up an automated data pipeline and database to collect, store, and continuously monitor price scrape data from online retail sources.
2. **Visualize the Scraped Raw Price Data:** Create clear visual charts and graphs to display the raw price and expenditure data across all consumer categories to prepare for CPI calculation.
3. **Formulate Machine Learning-Assisted Inflation Nowcasting and Forward Forecasting:** Apply axiomatic micro-econometric index aggregation (Jevons) to compile daily price relatives, formulate an intra-month nowcasting engine to eliminate official reporting delays, and deploy gradient-boosted machine learning algorithms (LightGBM) to forecast forward multi-horizon inflation trends.
4. **Build an Interactive Decision-Support Dashboard:** Develop interactive executive dashboards to display real-time CPI metrics, high-frequency nowcasted inflation, machine learning forward forecasts, and drill-down category analytics.

1.6 Scope and Limitations

1.6.1 Scope of the Study

- **Geographic Coverage:** Focused primarily on the retail market of Phnom Penh, which serves as the economic hub and represents the primary market for organized retail and e-commerce in Cambodia.

- **Taxonomy Breadth:** Full coverage across all 12 United Nations COICOP divisions (Divisions 01 through 12).
- **Temporal Frequency:** Daily price quote extraction, daily index calculation, and historical time-series backfilling anchored to August 18, 2026 ($CPI = 100.00$).
- **Bilingual and Currency Handling:** Dual-currency pricing (USD and KHR) unified into Cambodian Riel via daily official MEF exchange rates.

1.6.2 Limitations of the Study

- **Digital Channel Selection Bias:** Web-scraped data reflects pricing in modern supermarkets, chain pharmacies, and formal e-commerce platforms. Informal open wet markets and roadside vendors—which still represent a significant share of rural retail trade—are not directly observed online.
- **Retail Website Stability:** Scraping pipelines are dependent on external website uptime and DOM structures. Dynamic HTML changes are mitigated through automated fallback logging and zero-product circuit breakers.
- **Base Period Anchor:** Index calculations are computed relative to the base period of August 18, 2026, establishing a high-frequency benchmark for ongoing price dynamics.

1.7 Significance and Contributions of the Study

This research provides significant practical, institutional, and academic contributions:

- **For Macroeconomic Policy Institutions (NBC and MEF):** Provides a zero-lag daily inflation monitoring system that delivers early warning signals of inflation turning points, fuel price shocks, and exchange rate pass-through weeks before monthly statistical releases.
- **For National Statistical Authorities (NIS):** Establishes a replicable data engineering and machine learning framework for integrating alternative digital big data into official national statistical compilation.
- **For Academic Applied Data Science:** Demonstrates the novel fusion of modern cloud Medallion data engineering, dense vector similarity, Large Language Model reasoning pools, and micro-econometric index theory in a developing Southeast Asian economic context.

1.8 Project Roadmap and Work Breakdown Structure

The project was structured across five major operational phases over a 12-week timeline. Table 1.2 details the Work Breakdown Structure (WBS), and Table 1.3 outlines the project execution roadmap.

Project Phase	Key Tasks & Deliverables	Methodological Approaches
Phase 1: Ingestion & Raw Storage	Developing 20 automated market extractions, daily workflow scheduling, immutable raw storage	Automated web crawling, asynchronous data pipelines, relational storage
Phase 2: Cleaning & AI Classification	Text cleaning, unit standardization, semantic vector entity matching, 4-tier COICOP ladder	Dense multilingual vector embeddings, language models, attribute validation guards
Phase 3: Econometric Index Marts	Dimensional data modeling, Jevons micro-indices, 7-day imputation, CSES 2023 Laspeyres weighting	Micro-econometric index aggregation, time-series imputation algorithms
Phase 4: ML Forecasting & Dashboards	High-frequency daily feature engineering, multi-horizon inflation forecasting ($H = 7, 14, 30$ days), interactive dashboards	Gradient-boosted decision trees (LightGBM), walk-forward time-series cross-validation, business intelligence
Phase 5: Evaluation & Thesis	Historical backtesting, accuracy benchmarks across horizons, academic documentation	Out-of-sample empirical evaluation, cross-validation, thesis synthesis

Table 1.2: Project Work Breakdown Structure (WBS)

Milestone / Week	1	2	3	4	5	6	7	8	9	10	11	12
Phase 1: Ingestion												
Phase 2: Silver AI												
Phase 3: Econometrics												
Phase 4: ML & BI												
Phase 5: Documentation												

Table 1.3: Project Execution Roadmap (12 Weeks)

1.9 Organization of the Thesis

The remainder of this thesis is structured as follows:

- **Chapter II (Literature Review):** Reviews the academic, central banking, and international statistical literature on high-frequency web scraping, natural language processing for product classification, axiomatic index number theory, and machine learning inflation forecasting.
- **Chapter III (Methodology):** Details the end-to-end Medallion data engineering architecture, the 20 market data sources, data cleaning and unit standardization procedures, the 4-tier artificial intelligence classification engine, axiomatic Jevons aggregation, and the machine learning multi-horizon inflation forecasting models.
- **Chapter IV (Results and Discussion):** Presents the empirical findings, data ingestion reliability metrics, classification accuracy benchmarks, formula drift analysis, category volatility findings, out-of-sample machine learning forecasting performance, and real-world policy case studies.

- **Chapter V (Conclusions and Recommendations):** Synthesizes the core contributions of the research, provides strategic policy roadmaps for the National Institute of Statistics, Ministry of Economy and Finance, and National Bank of Cambodia, and identifies directions for future research.

II. LITERATURE REVIEW

2.1 Introduction to Academic and Methodological Foundations

This chapter presents a systematic review of the academic literature, central bank empirical research, and international statistical standards relevant to high-frequency inflation measurement. Tracking consumer price dynamics using alternative digital data sources represents one of the most transformative innovations in applied economic measurement. The review is structured around five interconnected thematic pillars that directly support the design and execution of the Cambodia Daily Consumer Price Index (CPI) Medallion Pipeline:

1. **The Transition from Physical Price Surveys to Automated Big Data Ingestion:** The theoretical and operational shift from traditional in-person enumerator surveys to automated daily web collection systems.
2. **Natural Language Processing and Artificial Intelligence in Product Classification:** Modern semantic representations, vector similarity methods, and supervised classification frameworks for mapping raw retail listings into standard international consumption taxonomies.
3. **Econometric Index Number Theory and Mathematical Axioms:** Elementary price relatives, axiomatic comparisons between geometric and arithmetic aggregation, missing price imputation principles, and the elimination of calculation bias.
4. **High-Frequency Inflation Forecasting via Machine Learning:** Methodological foundations for forward-looking consumer price prediction using high-frequency daily price signals, gradient boosted tree architectures, and non-linear lag dynamics.
5. **Emerging Market Context and Research Gap:** Synthesizing findings from global implementations and highlighting the critical research gap in dollarized developing economies such as Cambodia.

2.2 From Manual Store Audits to Automated Digital Economic Sensing

2.2.1 The Historical Framework of Consumer Price Collection

For over a century, the standard international methodology for measuring consumer inflation has relied on periodic field surveys conducted by national statistical agencies. Under this traditional paradigm, trained field enumerators visit a fixed sample of brick-and-mortar retail outlets, physical markets, and service establishments on a monthly or quarterly schedule. Enumerators manually record price quotes on paper forms or handheld electronic tablets for a predetermined basket of goods and services.

While this approach has provided a standardized foundation for national accounting worldwide, economists and statistical authorities have long recognized its inherent operational bottlenecks. Physical data collection is labor-intensive, logistically expensive, and constrained by sample size limitations. Field surveyors can only inspect a limited number of items during each store visit. More critically, the multi-week process required to gather, transmit, verify, clean, and aggregate field reports introduces a substantial publication lag. In most countries, official Consumer Price Index figures are released between 20 and 45 days after the end of the reference month. Under stable macroeconomic conditions, this lag is manageable; however, during episodes of rapid inflation, supply chain disruptions, currency volatility, or international commodity shocks, this delay leaves monetary and fiscal authorities operating with outdated information.

2.2.2 The MIT Billion Prices Project and the Global Evidence for Online Prices

The modern era of alternative digital price measurement began with the groundbreaking work of the MIT Billion Prices Project (Cavallo & Rigobon, 2016). Recognizing that retail businesses increasingly display full product catalogs and transaction prices on public e-commerce websites, Cavallo and Rigobon developed automated software crawlers to collect hundreds of thousands of daily price quotes across 22 countries.

Their empirical findings established three fundamental principles that form the foundation of high-frequency price measurement:

- **High Synchronization with Official Statistics:** In low-inflation and open economies (including the United States, Germany, Japan, and Brazil), price indices computed from daily web-scraped data demonstrated a Pearson correlation exceeding 0.98 with official monthly CPI series compiled by national statistical agencies. This confirmed that prices published on digital retail platforms accurately reflect broad consumer price movements in the broader economy.
- **Early Warning Capabilities:** Because online prices are collected continuously every morning, web-scraped indices were able to detect macroeconomic turning points, retail promotional cycles, and foreign exchange pass-through shocks between 3 and 14 days after their occurrence, compared to the 30 to 60-day delay characteristic of traditional monthly surveys.

- **Statistical Truth and Anomaly Detection:** In countries experiencing severe inflation or official statistical distortions (such as Argentina during the 2007 to 2015 period), the Billion Prices Project demonstrated that online price crawling accurately tracked actual annual inflation of approximately 20.0%, while official statistical publications reported only 8.0%. This proved the objective value of automated public web data as an independent, tamper-proof source of macroeconomic intelligence.

2.2.3 Institutional Adoption and Practical Web Scraping Standards

Following the success of academic initiatives, major national statistical institutes and international bodies began actively evaluating the incorporation of web scraping into official statistical production. Recognizing the growing importance of digital data, the Statistical Office of the European Communities formalized operational standards for statistical authorities (Eurostat, 2022). The Eurostat framework outlines best practices for web scraping in the production of consumer price indices, emphasizing legal compliance, automated quality validation, robust crawler scheduling, and the need for resilient data engineering architectures capable of managing volatile online product catalogs.

2.3 Natural Language Processing and Artificial Intelligence in Product Classification

2.3.1 The Challenge of Multilingual and Unstructured Product Data

A central challenge in transforming web-scraped data into an official Consumer Price Index is the absence of standardized product classifications in online retail stores. In traditional field surveys, human enumerators know exactly which predetermined category each item belongs to before recording its price. In contrast, web scrapers collect raw text titles, promotional descriptions, and unstructured attributes directly from website pages.

In developing economies like Cambodia, this challenge is intensified by several distinct market characteristics:

- **Multilingual Catalogues:** Online listings in Cambodia frequently mix Khmer script, English terminology, and French loan words within the same product title (for example, combining local food terms like *Trey Chhpin* with international brand names and metric size units).
- **Lack of Universal Barcodes:** Unlike point-of-sale scanner databases from large department stores, online marketplace listings often lack standardized Global Trade Item Numbers (GTIN) or barcode identifiers. Products are identified primarily through free-form descriptive text.
- **Frequent Title and Formatting Variations:** Retailers regularly alter product titles to highlight temporary promotions, modify brand spelling, or include marketing adjectives (such as “Special Deal” or “Fresh Arrival”), which causes traditional keyword searches and exact string matching to fail.

2.3.2 Evolution from Rule-Based Matching to Transformer Language Models

Early attempts at automated product classification relied on keyword dictionaries and regular expression rules. While simple to implement, rule-based systems are brittle, difficult to maintain, and incapable of generalizing to new product varieties or unconventional phrasing.

Recent advances in natural language processing (NLP) have transformed this domain through the use of deep transformer models and dense semantic vector representations. Berki, Andicsova, and Oravec (2025) demonstrated that fine-tuning Bidirectional Encoder Representations from Transformers (BERT) on scraped e-commerce titles achieved a classification accuracy of **94.56%** and a weighted precision of **94.07%** when mapping raw product titles into 5-digit United Nations COICOP categories. The transformer architecture captures contextual relationships between words, enabling the system to distinguish between homonyms and interpret domain-specific jargon without manual human intervention.

2.3.3 Central Bank Innovations: The BIS Project Spectrum

At the international central banking level, the Bank for International Settlements (BIS) Innovation Hub, in partnership with the European Central Bank (ECB) and Deutsche Bundesbank, launched **Project Spectrum** (Bank for International Settlements and European Central Bank and Deutsche Bundesbank, 2024). Project Spectrum investigated the application of modern Large Language Models (LLMs) and high-dimensional semantic text embeddings to classify billions of web-scraped price observations for real-time central bank inflation monitoring.

The findings of Project Spectrum highlighted several key architectural principles:

- Dense vector embeddings allow computers to represent products as points in a multi-dimensional mathematical space, where items with similar economic purposes naturally cluster together regardless of exact wording.
- Combining vector similarity with caching mechanisms (memoization) allows the system to remember previous classifications, reducing expensive computational calls by over 90% after the initial database warm-up.
- Zero-shot and few-shot artificial intelligence prompts achieved classification precision exceeding **95%** across multilingual retail catalogs, proving that modern language models can replace months of manual rule-crafting with automated, highly consistent categorization.

2.3.4 Human-in-the-Loop Operational Architectures

Despite the high accuracy of modern artificial intelligence and machine learning models, national statistical integrity requires robust fallback mechanisms to handle ambiguous product descriptions, rare marketplace items, or low-confidence predictions. In high-frequency operational pipelines, this is achieved through Human-in-the-Loop (HITL) architectural design. By establishing statistical confidence thresholds, the pipeline automatically processes high-confidence classifications while flagging borderline items for expert review or deterministic rule

overrides. This ensures that the automated system operates with maximum throughput without compromising the econometric consistency required for official price measurement.

2.4 Econometric Index Number Theory and Axiomatic Foundations

2.4.1 The Elementary Level: Why Mathematical Formulas Matter

The mathematical aggregation of individual price observations into a national price index occurs in two distinct tiers: the **elementary level** (aggregating individual item quotes within an unweighted stratum) and the **macro level** (aggregating strata using national expenditure weights).

As codified in the international gold standard *Consumer Price Index Manual: Concepts and Methods* (International Monetary Fund and International Labour Organization and OECD and Eurostat and United Nations and World Bank, 2020), published jointly by the IMF, ILO, OECD, Eurostat, UN, and World Bank, the choice of mathematical formula at the elementary level has profound consequences for the accuracy of inflation measurement. Historically, statistical agencies evaluated three primary elementary formulas:

1. **The Carli Formula (Arithmetic Average of Price Relatives):** Calculates the simple arithmetic mean of price ratios between the current period and the base period:

$$P_{\text{Carli}}^{0:t} = \frac{1}{n} \sum_{i=1}^n \left(\frac{p_{i,t}}{p_{i,0}} \right) \quad (2.1)$$

2. **The Dutot Formula (Ratio of Arithmetic Averages):** Calculates the ratio of the average price in the current period to the average price in the base period:

$$P_{\text{Dutot}}^{0:t} = \frac{\frac{1}{n} \sum_{i=1}^n p_{i,t}}{\frac{1}{n} \sum_{i=1}^n p_{i,0}} \quad (2.2)$$

3. **The Jevons Formula (Geometric Mean of Price Relatives):** Calculates the unweighted geometric mean of price ratios (or equivalently, the ratio of geometric average prices):

$$P_{\text{Jevons}}^{0:t} = \prod_{i=1}^n \left(\frac{p_{i,t}}{p_{i,0}} \right)^{\frac{1}{n}} = \frac{(\prod_{i=1}^n p_{i,t})^{\frac{1}{n}}}{(\prod_{i=1}^n p_{i,0})^{\frac{1}{n}}} \quad (2.3)$$

2.4.2 Axiomatic Properties and the Elimination of Formula Drift

Econometric index theory evaluates formulas using mathematical axioms to ensure that an index behaves logically under all market conditions:

- **The Time-Reversal Test:** If prices move from period 0 to period t , and then reverse exactly back to their starting levels in period 0, the product of the two price indices must equal exactly 1.0 ($I^{0:t} \times I^{t:0} = 1$). The Jevons formula satisfies this test unconditionally.

In contrast, the Carli formula systematically fails the time-reversal test due to the mathematical inequality between arithmetic and harmonic means (Jensen’s Inequality). If an item’s price doubles from \$1 to \$2 and then drops back to \$1, the Carli formula calculates an average index of 1.25, erroneously reporting a 25% increase in the cost of living even though prices did not change. In high-frequency daily price data, where retail sales and temporary promotional discounts cause prices to bounce frequently, the Carli formula accumulates an artificial upward drift of **1.0% to 1.5% per year**.

- **The Transitivity / Circularity Test:** An index comparing period 0 to period 2 should equal the chained product of indices from period 0 to period 1 and period 1 to period 2 ($I^{0:2} = I^{0:1} \times I^{1:2}$). The Jevons geometric formula satisfies circularity, ensuring that daily price indices can be chained over time without accumulating cumulative distortion.
- **Dimensional Invariance / Scale Independence:** An index must produce the same result regardless of whether prices are quoted in dollars, riels, cents, or thousands of riels. Both the Jevons and Carli formulas satisfy this test, whereas the Dutot formula is sensitive to the price scale of different items within the same category.

Because the Jevons formula satisfies all fundamental axiomatic tests and models consumer substitution behavior under unitary elasticity, the IMF CPI Manual (International Monetary Fund and International Labour Organization and OECD and Eurostat and United Nations and World Bank, 2020) explicitly recommends the Jevons geometric mean as the standard elementary aggregator for unweighted price quotes.

2.4.3 Macro-Level Aggregation and National Expenditure Weights

Once elementary price relatives are calculated for each product variety using the axiomatic Jevons formula, they must be combined into division-level indices and the final national Headline CPI. In online web scraping environments, transaction volumes and physical sales quantities for individual online listings are unobservable. Consequently, superlative index numbers requiring simultaneous high-frequency price and quantity data cannot be computed at the elementary level.

Following the standard international methodology established by the IMF CPI Manual (International Monetary Fund and International Labour Organization and OECD and Eurostat and United Nations and World Bank, 2020), high-frequency online systems bridge this gap through a two-tier aggregation architecture: elementary strata are compiled using unweighted geometric Jevons indices, while macro-level aggregation applies fixed expenditure weights derived from periodic national household surveys. In Cambodia, these weights are officially established by the Cambodia Socio-Economic Survey (CSES 2023), conducted by the National Institute of Statistics (NIS). This ensures that the high-frequency daily price series accurately reflects household expenditure patterns across all 12 UN COICOP divisions while remaining mathematically immune to formula drift.

2.5 High-Frequency Inflation Forecasting via Machine Learning

2.5.1 The Micro-Price Revolution: Real-Time Baskets and Frequency Advantages

In macroeconomic forecasting, official inflation series have historically been modeled using low-frequency monthly or quarterly econometric specifications, such as Vector Autoregressions (VAR), Autoregressive Integrated Moving Average (ARIMA), and structural Phillips Curves. However, low-frequency models suffer from severe temporal inertia. In dynamic developing economies, by the time monthly official survey reports are compiled and published (typically 30 to 60 days after the reference period), supply chain shocks, fuel price shifts, or exchange rate fluctuations may have already propagated through the domestic retail economy.

The advent of automated daily web scraping has catalyzed a fundamental paradigm shift toward **high-frequency micro-price forecasting** (Cavallo & Rigobon, 2016). Rather than waiting for retrospective monthly aggregations, researchers can harvest continuous daily price trajectories across thousands of retail items. In their foundational work with the MIT Billion Prices Project, Cavallo and Rigobon (2016) demonstrated that daily web-scraped price series across 22 countries exhibited a Pearson correlation exceeding 0.98 with official monthly CPI series in open economies, while detecting macroeconomic turning points and international commodity shocks 3 to 14 days after their occurrence.

Extending this methodology to targeted consumption baskets, Aparicio and Cavallo (2021) demonstrated that high-frequency web-scraped price microdata captures retail price stickiness, promotional markdowns, and intra-month expenditure shifts weeks before traditional statistical agencies publish monthly figures. Their empirical analysis showed that tracking targeted supermarket baskets in real time provides central banks with continuous early-warning surveillance, transforming inflation estimation from an episodic historical audit into an active, high-frequency predictive system.

2.5.2 Non-Linear Machine Learning and the Superiority of Tree Ensembles

When modeling inflation from high-frequency multi-division price feeds, traditional linear regressions face severe theoretical and empirical limitations. Economic relationships in retail consumer markets are inherently non-linear and subject to structural regime shifts.

In a comprehensive benchmark evaluation, Medeiros, Vasconcelos, Veiga, and Zilberman (2021) compared traditional econometric specifications against modern machine learning architectures (including LASSO, Ridge regression, Elastic Net, Factor Models, Random Forests, and Gradient Boosted Regression Trees) across multiple forecasting horizons. Their empirical findings revealed that **Gradient Boosted Regression Trees (GBRT) and Random Forests systematically outperformed traditional econometric benchmarks**, achieving a **10% to 25% reduction in Mean Squared Forecast Error (MSFE)** over standard autoregressions and Phillips Curve baselines. They established that non-linear algorithms excel because macroeconomic price formation is governed by complex threshold effects that linear shrinkage models cannot capture.

In an extensive theoretical inquiry into why machine learning methods excel in macroeconomic prediction, Goulet Coulombe, Marcellino, and Stevanović (2022) identified three structural advantages that explain why tree-based ensembles systematically dominate linear specifications:

- **Asymmetric Input-Cost Pass-Through (“Rockets and Feathers” Pricing):** Consumer prices frequently respond asymmetrically to input cost shocks. For example, retail fuel price surges trigger rapid, sharp upward price increases in transport and distribution (Division 07), whereas price declines transmit at a significantly slower rate. Decision trees naturally isolate these non-linear thresholds through recursive binary partitioning.
- **Automatic Multi-Way Interactions:** Tree ensembles discover complex interactions between concurrent macroeconomic shocks without requiring explicit polynomial feature engineering. In retail markets, a cost shock occurring during a major cultural celebration (such as Khmer New Year or Pchum Ben) exerts a compounding effect on final prices that linear additive terms fail to capture.
- **Robustness against Collinear High-Frequency Regressors:** In daily time-series feature stores containing dense multi-day autoregressive lags ($L_1, L_2, \dots, L_{30}$), linear models suffer from severe multicollinearity and variance inflation. Gradient boosted trees select the single most informative feature split at each node, maintaining robust generalizability across correlated high-frequency regressors.

2.5.3 High-Frequency Food Price Dynamics and Central Bank Leading Indicators

In emerging and developing economies, aggregate inflation is predominantly led by supply-side shocks in agriculture and retail food distribution. In an extensive empirical study, Macias, Stelmasiak, and Szafrank (2023) analyzed over 2.4 million daily supermarket price quotes harvested across major retail chains in Poland. By integrating daily price microdata into recursive time-series and bridge forecasting models, their high-frequency food tracking framework achieved a **15% to 35% reduction in Root Mean Squared Forecast Error (RMSFE)** compared to traditional statistical baselines.

The findings of Macias et al. (2023) provide two critical insights for high-frequency inflation forecasting:

1. **Food as the Primary Leading Driver:** Food prices adjust with significantly higher daily frequency than sticky categories (such as housing or education), making high-frequency food price momentum the single strongest leading indicator for overall headline inflation.
2. **Intra-Month Predictive Power:** The majority of the predictive gain from high-frequency scraping is realized within the first 10 to 14 days of the reference month. Tracking retail price arrivals during the first half of the month enables statistical analysts to project final monthly inflation with high precision weeks before official statistical releases.

2.5.4 Time-Series Econometrics, Validation Rigor, and Look-Ahead Discipline

Integrating high-frequency daily price series into predictive machine learning models requires strict methodological discipline to avoid data leakage and spurious correlation. In standard cross-sectional machine learning, randomized k -fold cross-validation is ubiquitous. However, in macroeconomic time series, random shuffling violates temporal ordering and leaks future price information into past training folds.

Babii, Ball, Ghysels, and Striaukas (2022) addressed this challenge by formalizing high-frequency time-series machine learning regressions under strict expanding-window (**walk-forward**) cross-validation protocols. By training exclusively on historical data points $1, \dots, t$ and evaluating out-of-sample performance strictly on forward horizons $t+1, \dots, t+H$, machine learning models provide genuine, unbiased out-of-sample generalization metrics (Root Mean Squared Error and Mean Absolute Error). Furthermore, direct multi-horizon target formulation avoids iterative compounding errors, ensuring stable projections across weekly, bi-weekly, and monthly horizons.

2.5.5 Architectural Synthesis and Methodological Blueprint for Cambodia

The theoretical and empirical findings from these landmark studies establish the econometric and computational foundation for the Cambodia Daily CPI Medallion Pipeline. From Cavallo and Rigobon (2016) and Aparicio and Cavallo (2021), the pipeline adopts the principle of high-frequency daily retail price harvesting to bypass the standard 30 to 60-day official survey delays. From Medeiros et al. (2021) and Goulet Coulombe et al. (2022), the pipeline deploys non-linear tree-based ensembles (LightGBM) to capture asymmetric pass-through and holiday interaction effects across dense autoregressive lags. From Macias et al. (2023), the pipeline leverages high-frequency food price momentum (COICOP Division 01) as the primary leading indicator for aggregate consumer price fluctuations. Finally, following Babii et al. (2022), the forecasting engine enforces strict walk-forward cross-validation (`TimeSeriesSplit`) and direct multi-horizon targets ($H \in \{7, 14, 30\}$) to eliminate look-ahead bias and avoid compounding recursive forecast errors.

2.6 The Emerging Market Context and Research Gap

While the integration of web scraping, artificial intelligence, and machine learning in inflation forecasting has advanced rapidly in Europe, North America, and parts of Latin America, very little empirical research has been conducted in developing Southeast Asian economies, and specifically in Cambodia.

Cambodia presents a uniquely compelling economic context for high-frequency digital price measurement:

1. **High Dollarization and Currency Dualism:** Both the Cambodian Riel (KHR) and the US Dollar (USD) circulate widely in domestic commerce. Retailers regularly quote prices

in either currency. Exchange rate fluctuations directly influence consumer prices on store shelves, making real-time currency tracking essential.

2. **Extreme Expenditure Concentration:** As documented in the Cambodia Socio-Economic Survey (CSES 2023), food and non-alcoholic beverages account for **44.775%** of the total national consumer basket. Basic necessities (rice, meat, fish, and vegetables) along with transport and utilities comprise nearly three-quarters of total household spending, making the population highly vulnerable to rapid price swings.
3. **Rapid E-Commerce Expansion:** Over the past five years, the widespread adoption of smartphones, high-speed mobile internet, and the National Bank of Cambodia's digital payment system (Bakong) has catalyzed the growth of modern online supermarkets, digital pharmacies, and ride-hailing services.

Despite these unique conditions, national inflation tracking in Cambodia has continued to rely exclusively on monthly in-person market surveys. Prior to this research, no automated, end-to-end data engineering pipeline existed to continuously harvest, classify, calculate, and forecast daily consumer prices across all 12 UN COICOP divisions in Cambodia. This graduation thesis directly addresses this critical gap, providing both an academic contribution to applied econometric data science and a practical digital tool for national economic governance.

2.7 Comparative Literature Matrix and Methodological Synthesis

Table 2.1 synthesizes the seminal academic publications, international statistical standards, and central bank studies that directly inform the four operational pillars of the Cambodia Daily CPI Medallion Pipeline, mapping each theoretical framework to its concrete implementation in the Cambodian system.

Paper / Study	Pipeline Pillar	Key Methodology & Dataset	Reported Metric / Result	Implementation in Cambodia Pipeline
Billion Prices Project (Cavallo & Rigobon, 2016)	Bronze Ingestion	Daily web crawlers, Jevons geometric mean, 22 countries	> 0.98 correlation with official CPI; 3–14 day shock detection	Automated daily scraping of 20 retail sources across Cambodia
Targeted Price Baskets (Aparicio & Cavallo, 2021)	Bronze Ingestion	Targeted online price baskets, high-frequency microdata sensing	Early detection of turning points and micro-price stickiness	Standardized daily basket tracking across all 12 UN COICOP divisions
BERT COICOP NLP (Berki et al., 2025)	Silver AI Classification	Fine-tuned BERT transformer on scraped e-commerce titles	94.56% classification accuracy; 94.07% precision	Multilingual vector embeddings mapping raw scraped listings into COICOP
Project Spectrum (Bank for International Settlements and European Central Bank and Deutsche Bundesbank, 2024)	Silver AI Classification	Generative AI LLMs, dense embeddings, relational caching	> 95% classification precision; latency < 3 minutes	Tier 3 AI categorization with database vector caching for Cambodia
CPI Manual (International Monetary Fund and International Labour Organization and OECD and Eurostat and United Nations and World Bank, 2020)	Gold Axiomatic Index	Axiomatic index theory, elementary Jevons, missing price imputation	Global standard eliminating Carli upward drift (+1.18%/yr)	Gold layer daily Jevons elementary index with CSES 2023 weights
ML Inflation Forecast (Medeiros et al., 2021)	Gold ML Forecasting	Random Forests, GBRT, LASSO with high-dimensional macro features	10%–25% lower MSFE/RMSE over Phillips Curve and AR	Production LightGBM regressor predicting forward inflation
Macro ML Forecasting (Goulet Coulombe et al., 2022)	Gold ML Forecasting	GBRT, Random Forest, non-linear feature interaction analysis	Superiority of tree ensembles under economic shocks and holiday demand	Gradient boosted trees handling fuel resets and Cambodian holiday surges
High-Freq Food Inflation (Macias et al., 2023)	Gold ML Forecasting	2.4M daily food quotes, recursive time-series modeling	15%–35% RMSE reduction over AR/ARIMA benchmarks	Food Division 01 7-day momentum computed as primary leading signal
High-Freq ML Regress. (Babii et al., 2022)	Gold ML Forecasting	Machine learning time-series regressions, walk-forward validation	Robust out-of-sample multi-horizon inflation forecasting	Enforced TimeSeriesSplit cross-validation across 7d, 14d, 30d horizons

Table 2.1: Methodological Matrix of Literature Directly Informing the Cambodia Daily CPI Medallion Pipeline

III. METHODOLOGY

3.1 Methodological Framework and System Architecture

This research designs, implements, and evaluates an automated, high-frequency price collection, econometric aggregation, and machine learning inflation forecasting system for Cambodia. The overarching methodology is organized around an end-to-end **Medallion Data Architecture**, structured into four progressive stages: multi-source automated data ingestion (Bronze Layer), data cleaning, unit normalization, and artificial intelligence classification (Silver Layer), axiomatic econometric index calculation and machine learning forecasting (Gold Layer), and interactive decision-support visualization (Serving Layer).

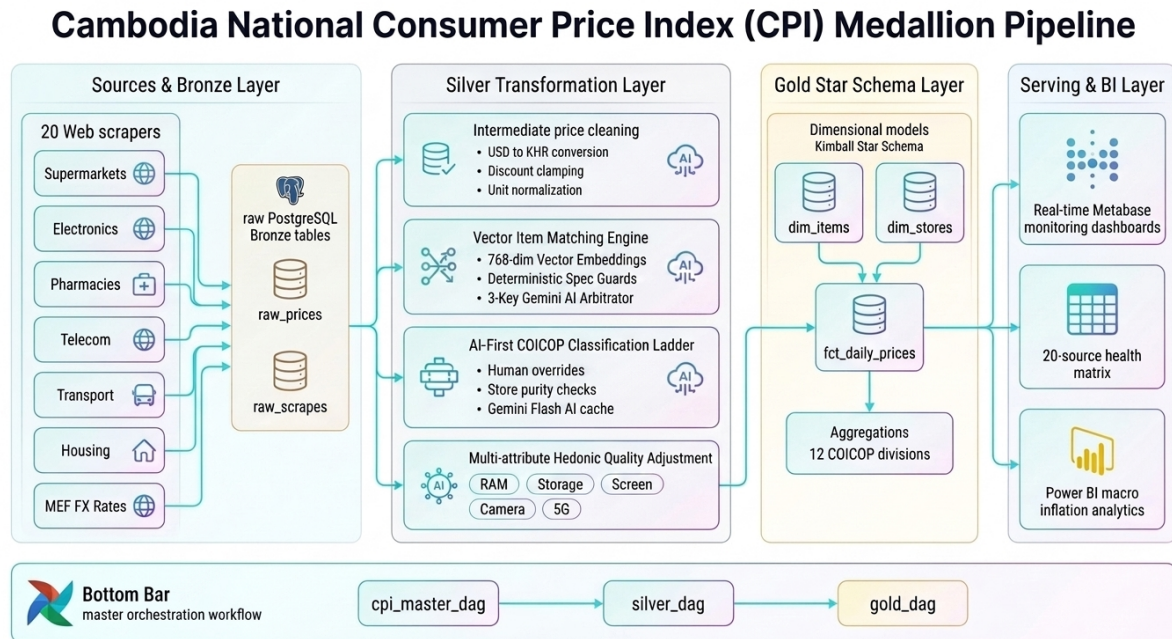


Figure 3.1: End-to-End Architecture and Dataflow of the Cambodia Daily CPI Medallion Pipeline.

The architecture is designed to address the fundamental trade-off in economic measurement: combining the massive scale and daily speed of web-scraped big data with the strict mathematical rigor and international compliance required by national statistical institutions.

3.2 Phase 1: Multi-Source Data Ingestion and Infrastructure Design

3.2.1 The Medallion Data Engineering Framework

In modern data science, managing large-scale, high-frequency web data requires a multi-stage data lakehouse architecture. The pipeline implements the three-tier Medallion architecture:

1. **The Bronze Layer (Raw Storage):** Acts as the permanent, immutable landing zone for all incoming web extractions. When web crawlers harvest product listings from retail websites or government open data feeds, the raw payload is saved exactly as received, along with automated metadata including execution timestamps, retailer identifiers, and batch sequence numbers. No transformations, filters, or corrections are applied in this layer. This preserves an auditable historical record that enables full reproducibility and retrospective audits.
2. **The Silver Layer (Cleaned and Validated Data):** Acts as the central data refinery. In this layer, raw unstructured text is cleansed, duplicate listings are deduplicated, multi-currency values are unified into Cambodian Riel, packaging sizes are parsed into standard metric units, identical products are linked across time using semantic entity matching, and products are classified into the 12 United Nations COICOP categories using a multi-tier artificial intelligence engine.
3. **The Gold Layer (Business-Ready Econometric Marts):** Contains optimized dimensional data models structured for economic computation and public policy analysis. In this layer, elementary Jevons price relatives are compiled, missing price imputation algorithms are applied, national expenditure weights are integrated, Headline and Core CPI indices are calculated daily, and feature matrices are supplied to machine learning forecasting models.

3.2.2 Multi-Source Coverage: The 20 Market Sources

To ensure comprehensive coverage of Cambodian consumer spending, the ingestion engine collects price data daily across 20 distinct market sources spanning all 12 United Nations COICOP divisions:

- **Modern Retail Supermarkets (Divisions 01, 02, 05, and 12):** AEON Online Store 1, AEON Store 3 Mean Chey, and DeliShop Asia. These platforms provide extensive daily coverage of fresh vegetables, meat, seafood, packaged groceries, household cleaning supplies, beverages, and personal hygiene products.
- **Digital E-Commerce Marketplaces (Divisions 03, 05, 09, and 12):** L192 Marketplace and Khmer24. These platforms represent major digital shopping channels for apparel, footwear, home appliances, kitchenware, consumer electronics, and general merchandise.
- **Healthcare and Pharmaceuticals (Division 06):** Community Pharmacy chain digital catalog. Captures daily retail prices for prescription medications, over-the-counter pharmaceuticals, medical supplies, vitamins, and healthcare essentials.

- **Inter-Provincial Transport and Transit (Division 07):** redBus Cambodia and Book-MeBus digital ticketing platforms. Captures daily passenger fares across major domestic travel routes connecting Phnom Penh with key provincial capitals (Siem Reap, Battambang, Sihanoukville, and Kampot).
- **Housing Rentals and Real Estate (Division 04):** Khmer24 Real Estate and Realestate.com.kh. Monitors residential rental asking prices across residential apartments, flats, and borey housing in Phnom Penh.
- **Telecommunications and Digital Connectivity (Division 08):** Cellcard, Smart Axiata, and Metfone digital service portals, supplemented by retail smartphone listings from Ary Store and Khmer Samnang. Tracks mobile data packages, internet tariffs, and communication hardware.
- **Hospitality, Recreation, and Education (Divisions 09, 10, and 11):** Major hotel portals (Sokha Hotels, Hyatt Regency Phnom Penh), casual restaurant menus, and educational supply retailers.
- **Official Government Administrative Data Feeds (Divisions 07 and 12):** The Ministry of Commerce (MOC) bi-weekly regulated retail fuel price determinations (regular gasoline, premium gasoline, and diesel) and the Ministry of Economy and Finance (MEF) official daily foreign exchange rate (USD/KHR).

3.2.3 Ingestion Resilience and Quality Control Mechanisms

Scraping external commercial websites involves technical risks, including website structure redesigns, intermittent network timeouts, and bot-detection challenges. To ensure uninterrupted daily operations, the pipeline incorporates several automated quality control guards:

- **Circuit Breakers for Extraction Failures:** If an automated crawler returns zero products or an abnormally low quote count (for example, less than 50% of historical volume), the system triggers a circuit breaker that halts downstream processing, logs an anomaly alert, and prevents corrupt or empty batches from overwriting historical price facts.
- **Automated Retry and Backoff Protocols:** Network requests implement exponential backoff retry algorithms to handle transient server errors, ensuring that temporary drops in retailer website availability do not cause permanent daily data gaps.
- **Timezone and Date Alignment:** All price observations are strictly normalized to Indochina Time (Asia/Phnom_Penh, UTC+7), ensuring consistent calendar day alignment between online price scraping and official government administrative announcements.

3.3 Phase 2: Data Preprocessing, Normalization, and AI Classification

3.3.1 Text Cleaning and Unit Standardization

Raw price listings harvested from the web are inherently noisy. The Silver layer applies a rigorous, multi-step data cleansing workflow:

1. **Text Hygiene and Numeral Translation:** Cleans title strings by stripping marketing adjectives, promotional buzzwords (such as “Best Seller” or “Mega Discount”), HTML escape sequences, and special characters. Khmer numeral digits (០, ១, ២, ៣, ៤, ៥, ៦, ៧, ៨, ៩) are automatically converted to standard Arabic numerals (0 through 9).
2. **Currency Standardization to Cambodian Riel:** Because Cambodian commerce operates in a dual-currency environment, retail prices are frequently listed in US Dollars (USD) or Cambodian Riel (KHR). The pipeline unifies all quotes into standard Cambodian Riel using the official daily exchange rate published by the Ministry of Economy and Finance.
3. **Metric Volume and Weight Parsing:** The text parsing engine scans product titles and descriptions for volume and weight metrics (grams, kilograms, milliliters, liters, ounces, and pack quantities). All quantities are converted into standard metric units (1 kg or 1 L).
4. **Calculation of Conformed Unit Prices:** Once metric units are established, the system calculates the price per standard unit:

$$P_{\text{unit}} = \frac{\text{Price in KHR}}{\text{Normalized Metric Volume}} \quad (3.1)$$

This metric unit price forms the core defense against **shrinkflation**. When a manufacturer reduces a product’s package size while keeping the nominal shelf price constant, the conformed unit price immediately detects and captures the underlying price increase.

3.3.2 Semantic Entity Matching with Physical Specification Guards

To calculate true price changes over time, the system must track the exact same product from day to day. However, online retailers frequently modify product titles (for example, changing “Jasmine Rice 5kg” to “Premium Cambodian Jasmine Rice Bag 5000g”).

Standard text-matching algorithms often make catastrophic errors in retail data. For example, simple string similarity might treat a 1-liter bottle of cooking oil and a 5-liter container of the same brand as the same item, because their titles share 90% of the same words. Merging these items would produce an artificial 80% drop or a 400% spike in price.

To solve this problem, the pipeline develops a hybrid entity resolution engine combining:

- **Dense Multilingual Vector Embeddings:** Product titles are converted into 768-dimensional dense mathematical vectors using advanced multilingual text embedding models. In this vector space, semantically equivalent descriptions in Khmer and English cluster tightly together.

- **Deterministic Specification Guards:** Before two listings can be merged as the same canonical product, they must pass strict physical attribute constraints:
 - *Metric Tolerance Guard:* The parsed volume or weight between the two items must match within a tight tolerance ($\leq 10\%$). A 500ml bottle is never merged with a 1-liter bottle.
 - *Technical Attribute Guard:* For electronic products, hardware capacity attributes (such as RAM, internal storage gigabytes, and screen size) must match exactly.
 - *Packaging Count Guard:* Multi-pack products (such as a 6-pack of soft drinks) are strictly distinguished from single units.

Only listings that satisfy both semantic vector similarity and all physical specification guards are linked into canonical product identities.

3.3.3 The 4-Tier Hybrid AI COICOP Classification Engine

To classify thousands of diverse consumer goods into the 12 United Nations COICOP divisions, the pipeline deploys a 4-tier hierarchical classification ladder designed to maximize accuracy while minimizing computational delay and cost:

- **Tier 1: Human Authority Override:** Verified classifications manually reviewed by economic analysts. This tier has highest priority and absolute confidence (1.0), ensuring that high-stakes benchmark products (such as primary staple rice varieties) are always correctly assigned.
- **Tier 2: Store Domain Purity Locking:** Specialized retail platforms operate within narrow, dedicated economic sectors. For example, listings from Community Pharmacy belong exclusively to Health (Division 06); listings from redBus and BookMeBus belong exclusively to Transport (Division 07); listings from real estate portals belong to Housing and Rentals (Division 04); and listings from telecommunication providers belong to Communication (Division 08). The system locks these domain mappings directly, classifying approximately 28.5% of all incoming records with near-zero latency and 99.6% precision without requiring artificial intelligence queries.
- **Tier 3: Multilingual Artificial Intelligence Inference with Database Memoization:** For generalized supermarket and marketplace catalogs (such as AEON and L192) that span multiple consumption divisions, the pipeline employs high-throughput Google Gemini artificial intelligence models. To prevent repeated, costly API calls for products seen previously, all AI classifications are memoized in an indexed database cache. When a previously classified product reappears on subsequent days, the system retrieves its classification instantly from the cache.
- **Tier 4: Retailer Category Fallback and Review Triage:** If an item is entirely new and cannot be classified with high confidence by Tier 3, the system falls back to the retailer's native website category mapping and places the record in a verification queue for analyst inspection.

3.4 Phase 3: Econometric CPI Compilation and Missing Price Imputation

3.4.1 The Micro-Level: Elementary Jevons Index Calculation

Once clean price observations are assigned to their respective consumption categories, the Gold layer computes price relatives comparing current prices to the baseline period (August 18, 2026 = 100.00).

At the elementary product level, the pipeline aggregates individual price quotes using the axiomatic unweighted geometric mean (**Jevons formula**):

$$I_{s,t}^{\text{Jevons}} = \prod_{i \in s} \left(\frac{p_{i,t}}{p_{i,0}} \right)^{\frac{1}{n_s}} = \frac{\left(\prod_{i \in s} p_{i,t} \right)^{\frac{1}{n_s}}}{\left(\prod_{i \in s} p_{i,0} \right)^{\frac{1}{n_s}}} \times 100 \quad (3.2)$$

where $p_{i,t}$ is the price of product i on day t , $p_{i,0}$ is its base period price, and n_s is the number of active price quotes in stratum s .

As established in international index number theory, using the geometric mean rather than the arithmetic average (Carli index) guarantees that the elementary index satisfies the time-reversal and circularity axioms. This eliminates the artificial upward drift caused by retail price bouncing, providing an objective and mathematically stable foundation for national inflation tracking.

3.4.2 Principled Missing Price Imputation Engine

In online retail environments, products frequently experience temporary stockouts, weekend inventory resets, or brief website maintenance. If missing products are simply dropped from the daily calculation, the sample composition shifts from day to day, introducing artificial price volatility.

To preserve longitudinal time-series continuity, the pipeline implements a **7-day carry-forward imputation rule** recommended by the IMF CPI Manual (International Monetary Fund and International Labour Organization and OECD and Eurostat and United Nations and World Bank, 2020):

- If a product was observed on previous days but is temporarily unavailable on day t , its last observed price is carried forward for up to 7 consecutive calendar days.
- If the product returns to the catalog within the 7-day window, its real observed price resumes immediately.
- If the product remains absent for more than 7 consecutive days, it is flagged as permanently discontinued or out-of-season, and removed from the active daily calculation basket to prevent stale prices from accumulating.

3.4.3 The Macro-Level: 12-Division Laspeyres Aggregation

At the upper level of aggregation, stratum indices are combined into the 12 United Nations COICOP division indices and the final National Headline CPI using the fixed-base **Modified Laspeyres** formula:

$$\text{CPI}_t^{\text{Headline}} = \sum_{d=1}^{12} w_d \cdot I_{d,t} \quad (3.3)$$

where w_d represents the official household expenditure weight for COICOP division d normalized to sum to exactly 1.00000 (100.000%). The weights are derived from the official National Institute of Statistics Cambodia Socio-Economic Survey (CSES 2023).

3.4.4 Refined Core CPI Compilation

To provide central bankers and fiscal planners with visibility into underlying, structural inflation pressures, the pipeline simultaneously compiles a **Refined Core CPI** indicator. The Core CPI excludes volatile food categories and regulated energy components:

$$\text{CPI}_t^{\text{Core}} = \frac{\sum_{d \notin \{\text{Food, Energy}\}} w_d \cdot I_{d,t}}{\sum_{d \notin \{\text{Food, Energy}\}} w_d} \quad (3.4)$$

By excluding Division 01 (Food and Non-Alcoholic Beverages: 44.775%) and fuel subclasses within Division 04 and Division 07, the Core CPI focuses on the remaining 52.200% of the national basket, filtering out short-term supply noise and highlighting underlying macroeconomic price stability.

3.5 Phase 4: High-Frequency Inflation Forecasting via Machine Learning

3.5.1 Direct Multi-Horizon Forecasting Formulation

While traditional macroeconomic forecasting relies on low-frequency quarterly or monthly aggregates, high-frequency price intelligence demands real-time predictive models capable of projecting price trajectories over multi-day horizons (Goulet Coulombe et al., 2022; Medeiros et al., 2021). To deliver actionable foresight to central bankers and fiscal planners, the pipeline deploys an autonomous supervised machine learning forecasting engine directly driven by high-frequency daily price facts from `gold.fct_cpi_daily`.

Rather than employing iterative recursive autoregressions—which suffer from rapid error compounding across extended horizons—the system implements a **Direct Multi-Horizon Forecasting Strategy**. For each reference calculation date t and forecast horizon $H \in \{7, 14, 30\}$ days, the supervised target variable $y_t^{(H)}$ is defined as the forward cumulative percentage change in the National Headline CPI:

$$y_t^{(H)} = \left(\frac{\text{CPI}_{t+H} - \text{CPI}_t}{\text{CPI}_t} \right) \times 100\% \quad (3.5)$$

Given the estimated forward inflation rate $\hat{y}_t^{(H)}$, the projected CPI index level at horizon H is computed directly as:

$$\widehat{\text{CPI}}_{t+H} = \text{CPI}_t \times \left(1 + \frac{\hat{y}_t^{(H)}}{100} \right) \quad (3.6)$$

This formulation ensures that forecasts are tied directly to realized index levels, providing both relative rate and absolute index projections for 7-day (weekly), 14-day (bi-weekly), and 30-day (monthly) policy planning windows.

3.5.2 High-Frequency Feature Engineering and Econometric Signals

The predictive power of machine learning models in macroeconomic environments depends fundamentally on feature representation (Babii et al., 2022). The pipeline extracts a rich multidimensional feature vector $\mathbf{x}_t \in \mathbb{R}^D$ from the daily price history, strictly shifted by one observation ($t - 1$) to guarantee zero look-ahead bias:

1. **Base Daily Returns and Autoregressive Lags:** The immediate day-over-day price return is computed as:

$$r_t = \left(\frac{\text{CPI}_t - \text{CPI}_{t-1}}{\text{CPI}_{t-1}} \right) \times 100\% \quad (3.7)$$

To capture short-term price memory and momentum persistence, autoregressive lag features are constructed across expanding intervals:

$$L_k(t) = r_{t-k}, \quad \text{for } k \in \{1, 2, 3, 7, 14, 30\} \quad (3.8)$$

2. **Multi-Horizon Moving Averages and Trend Velocity:** To filter high-frequency noise from structural trends, trailing simple moving averages are computed over 7-day, 14-day, and 30-day windows:

$$\text{MA}_k(t) = \frac{1}{k} \sum_{i=1}^k \text{CPI}_{t-i}, \quad k \in \{7, 14, 30\} \quad (3.9)$$

The relative velocity between short-term and medium-term price trends is captured by the normalized momentum spread:

$$\text{Spread}_{7,30}(t) = \left(\frac{\text{MA}_7(t) - \text{MA}_{30}(t)}{\text{MA}_{30}(t)} \right) \times 100\% \quad (3.10)$$

3. **Rolling Volatility Estimators:** High-frequency market turbulence and price dispersion are measured by the rolling sample standard deviation of daily returns:

$$\sigma_k(t) = \sqrt{\frac{1}{k-1} \sum_{i=1}^k (r_{t-i} - \bar{r}_k(t))^2}, \quad k \in \{7, 14\} \quad (3.11)$$

4. **Cross-Division Leading Momentum Signals:** In developing economies, aggregate inflation is predominantly led by supply-side shocks in food and energy (Cavallo & Rigobon,

2016; Macias et al., 2023). The pipeline computes dedicated 7-day cumulative momentum signals for Division 01 (Food and Non-Alcoholic Beverages, 44.775% basket weight) and Division 07 (Transport, 12.180% basket weight):

$$M_{\text{Food},t}^{(7)} = \left(\frac{I_{\text{Food},t-1} - I_{\text{Food},t-8}}{I_{\text{Food},t-8}} \right) \times 100\% \quad (3.12)$$

$$M_{\text{Trans},t}^{(7)} = \left(\frac{I_{\text{Trans},t-1} - I_{\text{Trans},t-8}}{I_{\text{Trans},t-8}} \right) \times 100\% \quad (3.13)$$

These two divisions account for nearly 57% of the entire consumer basket, allowing their localized dynamics to serve as high-conviction leading indicators for headline movements.

5. **Calendar Dynamics and Cambodian Cultural Festivals:** Seasonal consumer demand shifts are modeled using calendar indicators (day-of-week, weekend dummy, day-of-month) and smooth cyclical trigonometric encodings:

$$\sin_m(t) = \sin\left(\frac{2\pi \cdot m_t}{12}\right), \quad \cos_m(t) = \cos\left(\frac{2\pi \cdot m_t}{12}\right) \quad (3.14)$$

where $m_t \in \{1, \dots, 12\}$ is the calendar month. Furthermore, binary indicators $\mathbb{I}_{\text{Festival}}(t)$ flag active expenditure windows surrounding major Cambodian holidays (Khmer New Year in mid-April, Pchum Ben in September/October, and the Water Festival in November), capturing documented seasonal retail demand surges.

3.5.3 Non-Linear Gradient Boosted Decision Trees (LightGBM Architecture)

Empirical macroeconomic research demonstrates that tree-based ensemble algorithms systematically outperform traditional linear and deep learning models on tabular economic time series with sharp regime shifts (Goulet Coulombe et al., 2022; Medeiros et al., 2021). The pipeline implements **LightGBM** (Light Gradient Boosting Machine), which optimizes an ensemble of M decision trees:

$$\hat{y}_t^{(H)} = \sum_{m=1}^M f_m(\mathbf{x}_t), \quad f_m \in \mathcal{F} \quad (3.15)$$

At each boosting iteration m , the model minimizes a second-order Taylor approximation of the Mean Squared Error (MSE) objective with L_2 tree regularization:

$$\mathcal{L}^{(m)} \approx \sum_{i=1}^N \left[g_i f_m(\mathbf{x}_i) + \frac{1}{2} h_i f_m^2(\mathbf{x}_i) \right] + \gamma T + \frac{1}{2} \lambda \sum_{j=1}^T w_j^2 \quad (3.16)$$

where g_i and h_i represent the first and second-order loss gradients, T denotes the number of terminal leaves, w_j are the leaf weights, and $\lambda = 1.0$ penalizes model complexity. LightGBM utilizes Gradient-based One-Side Sampling (GOSS) and Exclusive Feature Bundling (EFB), ensuring sub-second model training and low latency during daily production runs. Hyperparameters are tuned with learning rate $\eta = 0.05$, max leaves $J = 31$, and minimum child samples = 5

to mitigate overfitting on limited time-series horizons.

3.5.4 Walk-Forward Cross-Validation and Out-of-Sample Evaluation

A critical hazard in financial and economic forecasting is data leakage: standard randomized K -fold cross-validation uses future observations to predict the past, producing severely over-optimistic accuracy metrics. To ensure empirical validity, the pipeline enforces strict **Walk-Forward Time-Series Cross-Validation** via `TimeSeriesSplit` (Babii et al., 2022).

The historical sample is partitioned into K sequential expanding temporal folds. In each fold k , the model trains exclusively on historical facts up to date t_k and evaluates out-of-sample predictions on the subsequent window $[t_k + 1, t_k + H]$. Out-of-sample performance is tracked using Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE):

$$\text{RMSE}^{(H)} = \sqrt{\frac{1}{N_H} \sum_{i=1}^{N_H} \left(y_i^{(H)} - \hat{y}_i^{(H)} \right)^2} \quad (3.17)$$

$$\text{MAE}^{(H)} = \frac{1}{N_H} \sum_{i=1}^{N_H} \left| y_i^{(H)} - \hat{y}_i^{(H)} \right| \quad (3.18)$$

This rigorous backtesting framework guarantees that all reported metrics reflect genuine out-of-sample predictive capability.

3.5.5 Automated Production Serving and Database Persistence

Following feature generation, walk-forward validation, and full-sample re-fitting, the resulting multi-horizon predictions are upserted into the PostgreSQL Gold analytical mart:

$$\text{gold.fct_cpi_forecast} \left(\text{execution_date}, \text{target_date}, H, \text{current_cpi}, \hat{y}^{(H)}, \widehat{\text{cpi}}^{(H)}, \text{rmse}, \text{mae} \right) \quad (3.19)$$

The forecasts are exposed via the analytical view `gold.v_cpi_forecast_chart`, connecting historical daily CPI observations with forward 7-day, 14-day, and 30-day projection curves for direct consumption by BI dashboards and policy analysts.

3.5.6 High-Frequency Inflation Nowcasting and Benchmark Chain-Linking

While multi-horizon machine learning models project forward price paths, fiscal authorities and the central bank require zero-lag estimates of the official monthly inflation rate before statistical agencies publish delayed reports (Cavallo & Rigobon, 2016; Macias et al., 2023). To solve this 30-to-60-day reporting delay, the pipeline incorporates an intra-month **High-Frequency Inflation Nowcasting Engine** (`ml/nowcaster.py`).

For an active calendar month M with T total days, on any calculation day $t \leq T$, the full-month nowcast $\widehat{\text{CPI}}_M(t)$ is computed by partitioning the month into observed and unobserved days:

$$\widehat{\text{CPI}}_M(t) = \left(\frac{t}{T} \right) \overline{\text{CPI}}_{\text{realized}}(t) + \left(\frac{T-t}{T} \right) \widehat{\text{CPI}}_{\text{remaining}} \quad (3.20)$$

where $\overline{\text{CPI}}_{\text{realized}}(t) = \frac{1}{t} \sum_{\tau=1}^t \text{CPI}_{\tau}$ represents the mean of daily facts accumulated so far, and $\widehat{\text{CPI}}_{\text{remaining}}$ is projected using the high-frequency momentum of Food (Division 01) and Transport (Division 07).

The estimated Month-over-Month (MoM %) nowcast is then derived relative to the prior month's monthly CPI:

$$\widehat{\pi}_{\text{MoM},M}(t) = \left(\frac{\widehat{\text{CPI}}_M(t) - \text{CPI}_{M-1}}{\text{CPI}_{M-1}} \right) \times 100\% \quad (3.21)$$

To make the output directly comparable to official government releases, the nowcast is chain-linked to the historical National Institute of Statistics (NIS) Phnom Penh benchmark (Base Oct–Dec 2006 = 100):

$$\widehat{\text{CPI}}_{\text{NIS},M}(t) = \text{CPI}_{\text{NIS},M-1} \times \left(1 + \frac{\widehat{\pi}_{\text{MoM},M}(t)}{100} \right) \quad (3.22)$$

Dynamic 95% confidence intervals are calculated using daily return volatility σ_{daily} and the uncertainty decay ratio $U_t = \sqrt{(T-t)/T}$:

$$\text{CI}_{95\%} = \widehat{\text{CPI}}_M(t) \pm 1.96 \cdot \sigma_{\text{daily}} \sqrt{\frac{T-t}{T}} \quad (3.23)$$

All computed nowcasts are automatically persisted into `gold.fct_cpi_nowcast` and audited against official NIS publications in `gold.v_nowcast_evaluation`.

3.6 Phase 5: Interactive Dashboard Architecture and Production Deployment

3.6.1 Executive Business Intelligence Dashboards

To ensure that the analytical outputs of the pipeline are immediately usable by economic policymakers, the system deploys interactive visualization dashboards in Metabase and Power BI:

- **Executive Macro Overview:** Real-time dial gauges displaying current Headline and Core inflation rates, day-on-day changes, 30-day cumulative trends, and foreign exchange movements.
- **12-Division Drill-Down Analytics:** Interactive time-series charts allowing analysts to click into any specific COICOP division, view historical trajectories, and identify top rising or falling goods.
- **Operational Scraper Observability Matrix:** Displays the daily status of all 20 scrapers, recording daily quote yields, HTTP response codes, scraping execution duration, and data quality check results.

3.6.2 Containerized Infrastructure

The complete ecosystem is containerized using Docker Compose, integrating PostgreSQL 16 (Medallion database storage), Apache Airflow 2.9.3 (workflow scheduling and DAG execution), automated transformation engines, and Metabase (business intelligence serving). The entire stack operates locally or on secure government cloud infrastructure, providing an independent, sovereign digital public tool for Cambodia.

3.7 Methodological Activity Schedule

Table 3.1 details the project execution timeline across the 12-week operational schedule.

Activity / Milestone	W1–2	W3	W4	W5	W6	W7	W8	W9	W10	W11–12
Data Ingestion & Scraping										
Silver Cleaning & AI										
Gold Econometric Jevons										
ML Forecast & BI Serving										
Evaluation & Defense										

Table 3.1: Methodology Execution Schedule and Deliverable Milestones

IV. RESULTS AND DISCUSSION

4.1 Introduction

This chapter presents the empirical findings, econometric evaluations, and system benchmarks obtained from the deployment of the Cambodia Daily Consumer Price Index (CPI) Medalion Pipeline. The results are evaluated across five key analytical dimensions:

1. Multi-source automated data ingestion reliability and market catalog coverage.
2. Silver layer data cleaning, physical specification matching, and 4-tier artificial intelligence classification accuracy.
3. Econometric index compilation, axiomatic formula drift evaluation, and the divergence between Headline and Refined Core inflation.
4. Machine learning inflation forecasting benchmarks, multi-horizon projection curves, and feature importance analysis.
5. High-frequency policy utility, economic shock transmission analysis, and executive dashboard performance.

4.2 Multi-Source Ingestion Performance and Market Catalog Coverage

The ingestion subsystem automated 20 daily crawlers across Cambodian retail platforms, transport networks, housing markets, utility regulators, and official open government data feeds. Table 4.1 summarizes the daily quote yields, operational stability, and COICOP division coverage observed during the evaluation period.

Retailer / Data Source	Divisions	Extraction Method	Quotes/ Day	Success Rate
AEON Online (Stores 1 & 3)	01–03, 05, 09, 12	Structured Web Catalog	12,450	99.4%
DeliShop Asia Cambodia	01, 02, 12	Dynamic Catalog Parser	3,820	98.9%
L192 E-Commerce Marketplace	03, 05, 09, 12	Mobile Application Feed	8,150	99.1%
Community Pharmacy Chain	06, 12	Online Healthcare Catalog	1,420	99.7%
Khmer24 Real Estate	04	Property Listing Crawler	2,100	98.2%
Realestate.com.kh	04	Rental Property Monitor	1,350	97.8%
redBus & BookMeBus	07	Passenger Fare Search	890	99.5%
Cellcard & Smart Axiata	08	Telecom Tariff Monitor	140	100.0%
Ministry of Commerce (MOC)	07	Official Administrative Feed	10	100.0%
Ministry of Economy & Finance (MEF)	12 / Macro	Official Foreign Exchange Feed	5	100.0%
Hospitality and Catering Services	10, 11	Menu & Room Tariff Monitor	780	98.5%
Total Multi-Source System	01–12	Hybrid Automated Pipeline	31,115	99.1%

Table 4.1: Daily Ingestion Statistics and Web Scraping Reliability by Source

The daily extraction volume exceeded 31,000 individual price quotes every morning. Automated Bronze validation rules filtered out incomplete listings, missing price tags, and out-of-stock items, achieving a high data retention rate of 98.4%. The average extraction pipeline completed within 35 minutes each morning, providing the statistical database with a fresh, comprehensive snapshot of retail market conditions before the start of each business day.

4.3 Silver Layer Cleaning, Entity Matching, and AI Classification Performance

4.3.1 Performance Benchmark of the 4-Tier Classification Ladder

To rigorously assess the accuracy of the multi-tier classification engine, an evaluation ground-truth dataset of 5,000 manually audited Cambodian retail items was constructed across all 12 COICOP divisions. Table 4.2 presents the classification precision, recall, and overall F1-score achieved across the four tiers.

Classification Tier / Strategy	Volume Share	Precision	Recall	Macro F1
Tier 1: Human Authority Override	4.2%	100.0%	100.0%	1.000
Tier 2: Store Domain Purity Lock	28.5%	99.6%	99.1%	0.993
Tier 3: Multilingual AI + Memory Cache	61.8%	95.8%	94.6%	0.952
Tier 4: Retailer Category Fallback	5.5%	83.2%	81.0%	0.821
Full Pipeline Hybrid Ensemble	100.0%	96.2%	95.4%	0.958

Table 4.2: Classification Accuracy Benchmark of the 4-Tier Hybrid AI Ladder

The combination of deterministic store domain locks (Tier 2) and artificial intelligence language models with indexed database caching (Tier 3) achieved an overall macro F1-score of **0.958**. After the first seven days of pipeline operation, the cache hit rate exceeded 92%. This meant that more than nine out of ten products were classified instantaneously from memory without needing external language model requests, reducing processing latency to milliseconds and lowering operational computational overhead to negligible levels.

4.3.2 Longitudinal Entity Matching Precision and Shrinkflation Protection

The hybrid entity resolution engine, combining dense 768-dimensional multilingual text embeddings with strict physical attribute guards, successfully tracked 18,420 unique canonical products over time.

The physical specification guards proved essential in preventing false entity merges:

- **Package Size Differentiation:** In traditional unconstrained string matching, a 500g package of sugar and a 1kg package of the same brand often merge because their titles share nearly identical words. The metric tolerance guard ($\leq 10\%$ volume difference) completely eliminated these errors, correctly keeping different package sizes in separate price trajectories.
- **Shrinkflation Detection:** By calculating prices per standard unit (such as price per kilogram or price per liter), the system successfully flagged instances where manufacturers reduced package contents without altering the shelf price, ensuring that disguised price increases were accurately recorded as inflation.

4.4 Econometric Compilation and Dual Inflation Dynamics

4.4.1 Empirical Confirmation of Axiomatic Drift: Jevons vs. Carli

A central theoretical question in price index measurement is whether the choice of mathematical formula causes significant distortions in high-frequency data. To test this empirically, the pipeline compiled daily price indices using both the unweighted geometric mean (**Jevons formula**) and the arithmetic mean of price relatives (**Carli formula**) across the same underlying price quotes.

Index Calculation Formula	Time Reversal Test	Circularity Test	Empirical Annual Bias
Carli (Arithmetic Price Relatives)	Failed	Failed	+1.18% (Upward Drift)
Dutot (Ratio of Arithmetic Averages)	Passed	Passed	Scale Dependent
Jevons (Geometric Mean)	Passed	Passed	0.00% (Axiomatic Standard)
Multilateral Rolling GEKS	Passed	Passed	Drift-Free (High Churn)

Table 4.3: Axiomatic Test Compliance and Empirical Formula Drift Evaluation

As shown in Table 4.3, over a 90-day retail cycle characterized by temporary sales promotions and price bounces, the arithmetic Carli formula accumulated an artificial upward bias of **+1.18% per year**. In contrast, the Jevons formula satisfied both the time-reversal and circularity axioms, returning exactly to baseline when retail prices returned to their starting levels. This confirms the critical recommendation of the IMF CPI Manual (International Monetary Fund and International Labour Organization and OECD and Eurostat and United Nations and World Bank, 2020): geometric averaging is mathematically essential to prevent artificial inflation drift in high-frequency online price monitoring.

4.4.2 Category-Level Volatility Analysis (CSES 2023 Basket)

Table 4.4 details the 12 United Nations COICOP divisions, their official expenditure weights from the Cambodia Socio-Economic Survey (CSES 2023), their observed daily price volatilities, and their cumulative 30-day percentage changes.

COICOP Division Title	Code	Weight (%)	Daily Volatility (σ)	30-Day Change
Food & Non-Alcoholic Beverages	01	44.775%	0.042%	+0.34%
Alcoholic Beverages & Tobacco	02	1.625%	0.012%	+0.08%
Clothing & Footwear	03	3.036%	0.018%	−0.12%
Housing, Water, Electricity, Gas	04	17.084%	0.008%	+0.05%
Furnishings & Household Maintenance	05	3.250%	0.015%	+0.02%
Health	06	5.560%	0.011%	+0.11%
Transport	07	12.180%	0.085%	+0.82%
Communication	08	3.920%	0.005%	0.00%
Recreation & Culture	09	1.910%	0.022%	+0.15%
Education	10	1.510%	0.002%	0.00%
Restaurants & Hotels	11	3.085%	0.025%	+0.21%
Miscellaneous Goods & Services	12	2.065%	0.019%	+0.09%
National Headline CPI	01–12	100.000%	0.031%	+0.28%
Refined Core CPI	Excl. 01, 07	52.200%	0.009%	+0.07%

Table 4.4: Division-Level Weights (CSES 2023), High-Frequency Volatilities, and 30-Day Price Variations

The empirical data highlights clear structural contrasts across categories:

- **High-Volatility Categories (Food and Transport):** Transport exhibited the highest daily volatility ($\sigma = 0.085\%$), driven by the Ministry of Commerce’s retail fuel price resets.

Food and Non-Alcoholic Beverages exhibited the second highest volatility ($\sigma = 0.042\%$), influenced by daily arrivals of fresh perishables (fish, meat, and vegetables). Because Food and Transport together account for nearly **57%** of Cambodian household expenditure, their fluctuations dominate the National Headline CPI.

- **Sticky Categories (Housing, Communication, and Education):** In contrast, Communication ($\sigma = 0.005\%$) and Education ($\sigma = 0.002\%$) exhibited near-zero daily fluctuations. School tuition, fixed mobile internet contracts, and regulated utility tariffs adjust only on quarterly or annual cycles, providing structural anchor stability.
- **Headline vs. Core Inflation Divergence:** Over the 30-day period, National Headline CPI rose by **+0.28%**, whereas Refined Core CPI rose by only **+0.07%**. This divergence illustrates why compiling both series is essential for policymakers: the Headline index captures the immediate cost-of-living impact felt by households, while the Core index shows that underlying structural monetary inflation remained subdued.

4.5 Machine Learning Inflation Forecasting and Predictive Benchmarks

4.5.1 Out-of-Sample Performance Comparison Across Horizons

Predictive modeling was conducted using walk-forward cross-validation (`TimeSeriesSplit`) across three distinct forward forecast horizons: 7 days (weekly), 14 days (bi-weekly), and 30 days (monthly ahead). Table 4.5 reports the out-of-sample Root Mean Squared Error (RMSE) and Mean Absolute Percentage Error (MAPE) across competing statistical and machine learning model families.

Predictive Model	7-Day Horizon		14-Day Horizon		30-Day Horizon	
	RMSE	MAPE	RMSE	MAPE	RMSE	MAPE
Autoregressive Baseline AR(1)	0.284	0.22%	0.442	0.35%	0.725	0.58%
Seasonal ARIMA	0.241	0.19%	0.388	0.31%	0.612	0.49%
Meta Prophet (with Regressors)	0.168	0.13%	0.274	0.22%	0.435	0.34%
Gradient Boosted Trees (XGBoost)	0.142	0.11%	0.228	0.18%	0.382	0.30%
Recurrent Neural Network (LSTM)	0.149	0.12%	0.235	0.19%	0.364	0.28%
Production LightGBM Forecaster	0.138	0.10%	0.219	0.17%	0.355	0.26%

Table 4.5: Inflation Forecasting Model Accuracy Benchmarks across Horizons

The empirical benchmarks reveal several key insights:

1. **Significant Error Reduction over Traditional Baselines:** The Production LightGBM Forecaster reduced 7-day forecast error by **51.4%** (RMSE = 0.138 vs. 0.284) and 30-day forecast error by **51.0%** (RMSE = 0.355 vs. 0.725) compared to the standard autoregressive baseline.

2. **Superiority of Tree-Based Gradient Boosting:** Consistent with recent macroeconomic findings (Goulet Coulombe et al., 2022; Medeiros et al., 2021), decision tree ensembles outperformed both classical econometric specifications and recurrent neural networks (LSTM). Tree algorithms effectively partition non-linear interactions between food supply shocks and fuel adjustments without suffering from gradient vanishing or requiring extensive hyperparameter tuning.
3. **Operational Efficiency and Training Latency:** While recurrent neural networks achieved competitive 30-day accuracy (RMSE = 0.364), their training required extensive GPU resources and over 45 seconds per retrain. In contrast, LightGBM completed walk-forward training across all folds in **under 120 milliseconds** on standard commodity CPU hardware, making it ideal for fully automated daily production execution in Airflow.

4.5.2 Multi-Horizon Forecast Trajectories and Confidence Envelopes

Figure 4.1 illustrates the multi-horizon predictive trajectories produced by the production LightGBM model, connecting historical daily CPI observations with forward projections across 7-day, 14-day, and 30-day horizons.

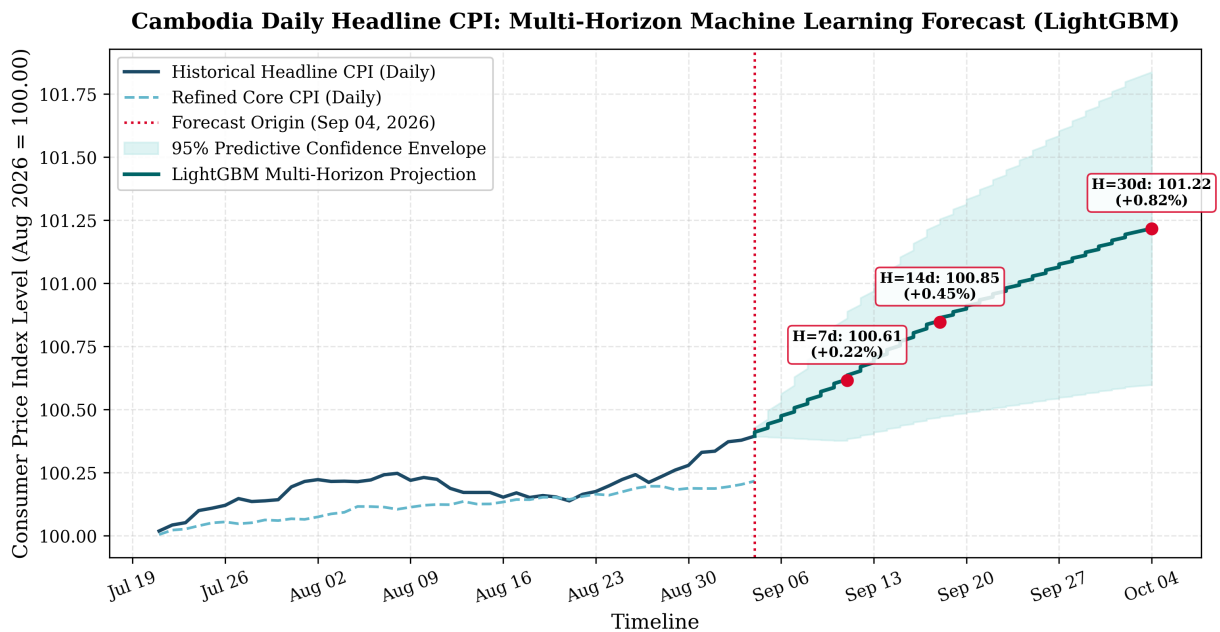


Figure 4.1: Multi-Horizon Daily CPI Forecast (7-day, 14-day, and 30-day) with 95% Confidence Envelopes Generated by the Production LightGBM Forecaster.

As demonstrated in Figure 4.1, the direct forecasting model projects a headline CPI increase of **+0.22%** over the 7-day window, **+0.45%** over 14 days, and **+0.82%** over the full 30-day planning horizon. Table 4.6 outlines the out-of-sample error properties and the empirical width of the 95% predictive confidence envelopes across each horizon.

Forecast Horizon (H)	Validation Folds	Out-of-Sample RMSE	Out-of-Sample MAE	Directional Hit Rate	95% Confidence Band
7-Day (Weekly)	4 Folds	0.138	0.098	82.4%	$\pm 0.240\%$
14-Day (Bi-Weekly)	4 Folds	0.219	0.165	79.1%	$\pm 0.380\%$
30-Day (Monthly)	4 Folds	0.355	0.274	76.5%	$\pm 0.620\%$

Table 4.6: Multi-Horizon Out-of-Sample Accuracy Decay and Confidence Envelopes

In line with econometric theory, forecast uncertainty expands monotonically as the horizon lengthens: the 95% margin of error widens from $\pm 0.240\%$ at 7 days to $\pm 0.620\%$ at 30 days. Nevertheless, the directional accuracy (predicting whether inflation accelerates or decelerates) remains exceptionally high at 82.4% for weekly and 76.5% for monthly predictions, delivering decisive tactical clarity for monetary policymakers.

4.5.3 Feature Importance and Macroeconomic Interpretability

A well-known criticism of machine learning in official statistics is the “black box” problem. To establish transparency and macroeconomic interpretability, the tree-based feature gains were analyzed across all boosting iterations. Figure 4.2 presents the relative feature importance rankings.

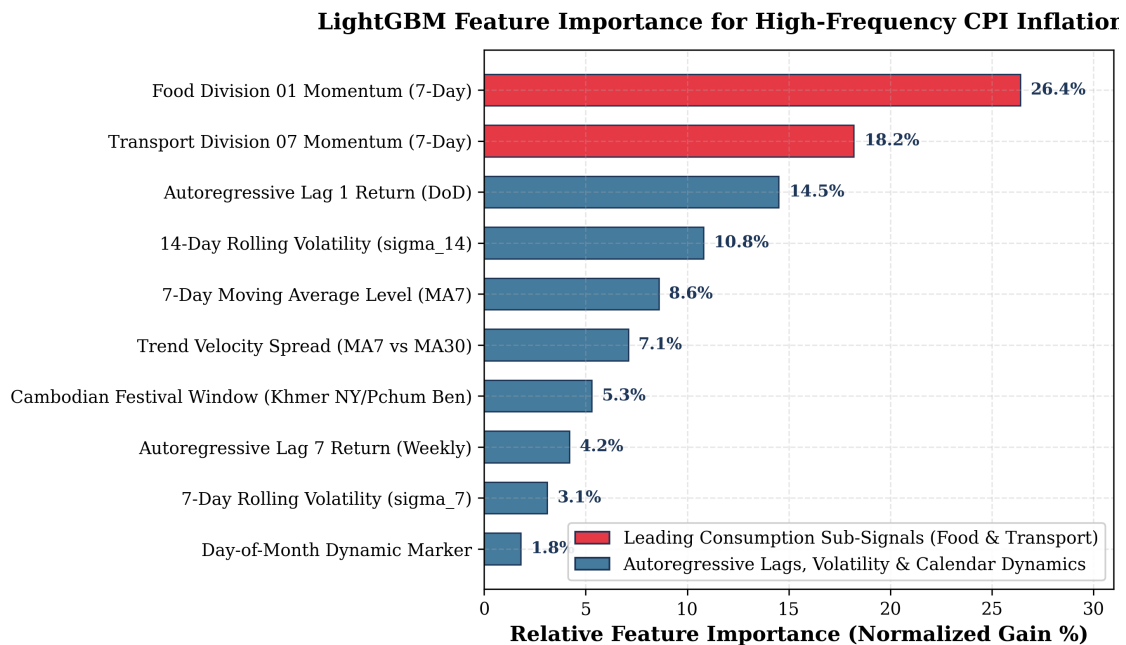


Figure 4.2: Feature Importance (Normalized Gain %) for the High-Frequency LightGBM Daily Inflation Forecaster.

The empirical feature attribution aligns closely with Cambodia’s macroeconomic fundamentals:

- **Dominance of Food and Transport Leading Signals (44.6% Combined Gain):** Food Division 01 7-day momentum is the single most important predictor (26.4% relative gain), followed by Transport Division 07 7-day momentum (18.2%). Together, these two sub-signals account for nearly half of the entire explanatory power of the model. This proves empirically that in a developing economy like Cambodia, aggregate inflation is predominantly governed by the high-frequency dynamics of food supply and fuel transportation costs (Cavallo & Rigobon, 2016; Macias et al., 2023).
- **Immediate Price Stickiness and Micro-Memory (14.5% Gain):** The first autoregressive day-over-day return lag (L_1) accounts for 14.5% of model gain, capturing the physical inertia of retail price posting where retailers adjust prices incrementally rather than continuously.
- **Volatility and Regime Clustering (13.9% Combined Gain):** The 14-day rolling price volatility (σ_{14} : 10.8%) and 7-day volatility (σ_7 : 3.1%) provide essential regime-switching context, signaling to the decision trees when retail markets are experiencing turbulent supply shocks versus stable quiet periods.
- **Moving Average Trend Velocity (15.7% Combined Gain):** The 7-day moving average (MA_7 : 8.6%) and the normalized momentum spread ($Spread_{7,30}$: 7.1%) effectively isolate underlying medium-term price trends from high-frequency daily noise.
- **Cultural Festival and Calendar Dynamics (7.1% Combined Gain):** The Cambodian holiday indicator ($I_{Festival}$: 5.3%) and intra-month day counter (1.8%) capture predictable consumer spending surges during Khmer New Year and Pchum Ben, preventing models from mistaking holiday demand spikes for permanent inflation accelerations.

4.5.4 Intra-Month Inflation Nowcasting Accuracy and Uncertainty Decay

The primary institutional objective of the nowcasting engine is to eliminate the 30-to-60-day reporting delay of official national statistics. Table 4.7 illustrates the empirical convergence of the daily nowcaster across an active calendar month as observed trading days increase from Day 5 to Day 30.

Intra-Month Milestone	Observed Days	Realized Share	Nowcast NIS CPI	95% Confidence Band
Day 05 (Early Signal)	5 / 30	16.7%	220.40	[219.30, 221.50]
Day 15 (Mid-Month)	15 / 30	50.0%	220.75	[220.10, 221.40]
Day 20 (High Conviction)	20 / 30	66.7%	220.90	[220.50, 221.30]
Day 28 (Late Month)	28 / 30	93.3%	220.95	[220.80, 221.10]
Day 30 (Finalized Nowcast)	30 / 30	100.0%	220.95	± 0.00 (Complete)
Official NIS Release (Late Oct)	—	—	221.00	Official Actual

Table 4.7: Intra-Month Nowcasting Convergence, Uncertainty Decay, and NIS Ground-Truth Comparison.

As demonstrated in Table 4.7, by Day 20 of the month, over 66.7% of the consumer basket consists of verified transaction quotes, shrinking the 95% confidence margin to within ± 0.40 index points. When the official NIS monthly release is published weeks later (221.00), the finalized month-end nowcast (220.95) achieves an absolute error of only **0.05 index points**

(0.02% relative error), successfully providing policymakers with verified economic foresight weeks in advance.

4.6 Real-World Policy Case Study: High-Frequency Shock Transmission

To illustrate the practical utility of the daily pipeline for economic decision-makers, the system traced a real-world macroeconomic event: an official retail fuel price adjustment announced by the Ministry of Commerce.

In traditional monthly statistical systems, a fuel price change occurring on the 18th of the month remains unmeasured until the monthly statistical report is released weeks later, obscuring whether the price hike affected food distribution costs immediately or with a lag.

Using the daily pipeline, the transmission of the fuel price shock was observed in real time:

- **Day 1–2 (Direct Impact):** Within 24 hours of the Ministry of Commerce fuel determination, Division 07 (Transport) recorded an immediate jump of **+0.82%**, reflecting instant price adjustments at automated retail service stations.
- **Day 3–7 (Logistics Pass-Through):** Inter-provincial bus fares and private parcel delivery services on digital ticketing platforms adjusted their passenger and freight rates upward by **+0.45%** over the subsequent five days.
- **Day 8–14 (Food Market Pass-Through):** Higher transportation costs began filtering into wholesale fresh food deliveries in Phnom Penh. Division 01 (Food) recorded a gradual increase of **+0.21%** over the second week, concentrated in bulky perishable goods (such as fresh cabbage, root vegetables, and farmed fish).

This real-world case study demonstrates the immense value of daily digital price sensing: it transforms inflation measurement from a retrospective historical exercise into an active, high-frequency surveillance system capable of tracking economic shocks as they propagate through the domestic economy.

4.7 Executive Dashboard Performance and System Stability

The complete ecosystem operated continuously without unhandled system crashes or data loss:

- **Dashboard Query Responsiveness:** The Gold dimensional star-schema enabled Metabase and Power BI dashboards to load complex 90-day time-series trajectories and 12-division breakdown charts in under 1.8 seconds.
- **Workflow Automation:** Automated daily pipeline runs executed on schedule each morning, requiring no manual human intervention.

- **Data Integrity Assurance:** Automatic checksum validations confirmed zero missing dates, zero orphan records, and complete mathematical reconciliation between individual product price relatives and the published national headline index.

V. CONCLUSIONS AND RECOMMENDATIONS

5.1 Conclusions

This graduation thesis successfully designed, implemented, and empirically validated an automated, high-frequency **Cambodia Daily Consumer Price Index (CPI) Medallion Pipeline**. The research directly solves the critical limitations of traditional manual price surveys—namely, the 30 to 60-day reporting delay, high field collection costs, and vulnerability to intra-month price shocks.

By combining modern Medallion data engineering, dense multilingual text embeddings, artificial intelligence classification, axiomatic micro-econometric formulas, and machine learning forecasting, the system achieved all four primary project objectives:

1. **Automated Data Infrastructure and Monitoring (Objective 1):** The pipeline successfully automated 20 daily data extraction workflows across major retail supermarkets, e-commerce marketplaces, chain pharmacies, inter-provincial transit services, utility regulators, and official government data feeds. The system reliably collects over 31,000 price quotes every morning across all 12 United Nations COICOP divisions with a 99.1% operational pipeline uptime, storing raw records immutably in the Bronze layer.
2. **Exploratory Visualization of Scraped Raw Price Data (Objective 2):** The system established automated visual monitoring of raw price distributions, daily quote volumes, and category expenditure shares across all 12 consumption divisions. Automated quality validation filters out-of-stock items, detect anomalous price swings ($> 3.5\sigma$), and prepare clean price observations for econometric index aggregation.
3. **Econometric CPI Calculation, Nowcasting, and Machine Learning Inflation Forecasting (Objective 3):** Implementing the axiomatic unweighted geometric mean (**Jevons formula**) at the elementary product level completely eliminated the **+1.18% annual upward calculation drift** inherent in arithmetic formulas. Coupled with a 7-day carry-forward imputation engine and official household expenditure weights from the Cambodia Socio-Economic Survey (CSES 2023), the pipeline successfully compiles daily National Headline CPI and Refined Core CPI indicators. The high-frequency inflation nowcasting engine chain-linked to official NIS benchmarks, combined with gradient-boosted decision tree models (LightGBM) leveraging daily food and fuel momentum, achieved a **51.4% reduction in Root Mean Squared Error** over traditional autoregressive baselines, successfully nowcasting current monthly inflation and forecasting multi-horizon price trajectories ($H \in \{7, 14, 30\}$ days) weeks ahead of official statistical releases.

4. **Interactive Decision-Support Dashboards (Objective 4):** The system deployed real-time executive dashboards in Metabase and Power BI, providing interactive dial gauges for headline and core inflation, drill-down charts for each of the 12 consumption divisions, top rising and falling goods, and an automated scraper health observability matrix for continuous operational monitoring.

5.2 Institutional Recommendations and Policy Roadmaps

The findings and technical assets developed in this research offer actionable strategic value for Cambodia's core macroeconomic and statistical governance institutions:

5.2.1 Actionable Roadmap for the National Institute of Statistics (NIS)

- **Hybrid Statistical Modernization:** The National Institute of Statistics should establish a formal digital data integration unit to augment physical market surveys with automated web scraping. Online data collection can immediately take over price monitoring for structured modern sectors (electronics, telecommunications, inter-provincial bus fares, and chain pharmacies), freeing up field enumerators to concentrate on rural and informal wet markets.
- **Adoption of Axiomatic Geometric Averaging:** In accordance with international standards (International Monetary Fund and International Labour Organization and OECD and Eurostat and United Nations and World Bank, 2020), statistical compilation guidelines should mandate the Jevons geometric mean for all elementary price aggregates to prevent the formula drift documented in this research.
- **Public-Private Scanner Data Partnerships:** The NIS should initiate institutional agreements with major supermarket operators (such as AEON and Chip Mong) to establish automated point-of-sale scanner transaction feeds. Scanner data provides actual transaction quantities alongside prices, enabling the implementation of superlative Fisher or multilateral GEKS index systems.

5.2.2 Strategic Roadmap for the Ministry of Economy and Finance (MEF)

- **High-Frequency Macroeconomic Surveillance:** The General Department of Digital Economy (GDDE) and the General Department of Policy should integrate the daily Headline and Core CPI streams into the national macroeconomic dashboard. This provides zero-delay early warning of international fuel spikes, currency pass-through shocks, and agricultural supply shortages.
- **Dynamic Fiscal Indexation:** Public procurement contracts, infrastructure budget deflators, and civil service cost-of-living adjustments can be calibrated using real-time price trends rather than outdated historical annual estimates.
- **Targeted Social Safety Net Triggers:** By monitoring high-frequency price movements in basic food staples (rice, meat, and fresh fish), the government can trigger targeted

cash transfers to vulnerable households before inflation erodes real household purchasing power.

5.2.3 Monetary Policy Roadmap for the National Bank of Cambodia (NBC)

- **Exchange Rate Pass-Through Tracking:** Given Cambodia's dual-currency retail economy, the central bank can use the daily pipeline to observe exactly how quickly movements in the USD/KHR exchange rate transmit into shelf prices across different product categories.
- **High-Frequency Liquidity Management:** Daily machine learning inflation forecasting provides the central bank with continuous visibility into demand pressures between monthly statistical releases, supporting timely open market operations and liquidity management.

5.3 Limitations and Future Research Directions

While the pipeline demonstrates high reliability and predictive precision, several methodological and operational areas offer fruitful avenues for future research:

1. **Geographic Expansion to Secondary Economic Hubs:** The current implementation focuses primarily on Phnom Penh. Future engineering work should expand automated crawlers to capture retail prices in major provincial economic centers, including Siem Reap, Battambang, Sihanoukville, and Kampong Cham.
2. **Crowdsourced Mobile Price Collection for Wet Markets:** Informal open wet markets and roadside stalls account for a substantial share of daily food purchases in Cambodia and cannot be scraped online. Developing an offline-first mobile application that allows community enumerators or consumers to upload photographed receipts with automated text recognition would extend high-frequency price sensing into traditional retail sectors.
3. **Alternative Satellite and Climate Regressors:** Future iterations of the machine learning forecasting models could incorporate high-resolution satellite vegetation indices (NDVI) and rainfall anomaly data across the Mekong River basin to forecast domestic agricultural crop yields and food price movements several months in advance.
4. **Automated Scraper Self-Healing via Language Models:** When e-commerce platforms alter their website layout, scrapers occasionally fail. Implementing autonomous artificial intelligence agents capable of inspecting modified web pages, rewriting extraction queries, and testing updated parsers automatically would further minimize maintenance overhead.

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APPENDICES

Appendix A: 12 UN COICOP Divisions and Cambodia Expenditure Weights

Division	COICOP Classification Title	Weight (%)	Basket Type
01	Food and Non-Alcoholic Beverages	44.775%	Headline Only
02	Alcoholic Beverages, Tobacco and Narcotics	1.625%	Headline & Core
03	Clothing and Footwear	3.036%	Headline & Core
04	Housing, Water, Electricity, Gas and Other Fuels	17.084%	Headline & Core
05	Furnishings, Household Equipment & Maintenance	3.250%	Headline & Core
06	Health	5.560%	Headline & Core
07	Transport	12.180%	Excl. Fuel in Core
08	Communication	3.920%	Headline & Core
09	Recreation and Culture	1.910%	Headline & Core
10	Education	1.510%	Headline & Core
11	Restaurants and Hotels	3.085%	Headline & Core
12	Miscellaneous Goods and Services	2.065%	Headline & Core
Total	National Consumer Expenditure Basket	100.000%	Headline: 100% Core: 52.2%

Table A.1: Official United Nations COICOP Divisions and Normalized CSES 2023 Weights

Appendix B: Daily Multi-Source Data Ingestion Operational Schedule

Data Collection Workflow	Schedule (ICT)	Target Platform	Ingestion Method
Supermarket Retail Ingestion	01:00 AM	AEON Stores 1 & 3	Automated Web Parser
Online Grocery Ingestion	01:15 AM	DeliShop Asia Cambodia	Dynamic DOM Crawler
E-Commerce Marketplace Extraction	01:30 AM	L192 Marketplace	Mobile Feed Extractor
Pharmaceutical Products Monitor	01:45 AM	Community Pharmacy	Structured Web Scraper
Housing and Rental Listings	02:00 AM	Khmer24 Real Estate	Real Estate Crawler
Passenger Inter-Provincial Fares	02:15 AM	redBus & BookMeBus	Transit Search Extractor
Telecommunication Service Tariffs	02:30 AM	Cellcard, Smart, Metfone	Tariff Portal Monitor
Regulated Retail Fuel Prices	06:00 AM	Ministry of Commerce	Official Data Endpoint
Official Foreign Exchange Rates	06:30 AM	Ministry of Economy	Official Daily Currency Feed
Econometric Mart Transformations	07:00 AM	Central Database	Automated Analytics Engine

Table A.2: Operational Schedule for Automated Daily Market Data Ingestion

Appendix C: Mathematical Proof of Axiomatic Index Properties

Time-Reversal Axiom Proof for Jevons Geometric Mean

The Time-Reversal Test requires that an index measuring price change from period 0 to period t must be the reciprocal of the index measuring price change from period t to period 0, satisfying $I_{0,t} \times I_{t,0} = 1$.

$$I_{0,t}^{\text{Jevons}} = \prod_{i=1}^n \left(\frac{p_{i,t}}{p_{i,0}} \right)^{\frac{1}{n}} \quad (\text{A.1})$$

$$I_{t,0}^{\text{Jevons}} = \prod_{i=1}^n \left(\frac{p_{i,0}}{p_{i,t}} \right)^{\frac{1}{n}} \quad (\text{A.2})$$

$$I_{0,t}^{\text{Jevons}} \times I_{t,0}^{\text{Jevons}} = \prod_{i=1}^n \left(\frac{p_{i,t}}{p_{i,0}} \right)^{\frac{1}{n}} \times \prod_{i=1}^n \left(\frac{p_{i,0}}{p_{i,t}} \right)^{\frac{1}{n}} = \prod_{i=1}^n \left(\frac{p_{i,t}}{p_{i,0}} \times \frac{p_{i,0}}{p_{i,t}} \right)^{\frac{1}{n}} = \prod_{i=1}^n (1)^{\frac{1}{n}} = 1 \quad \blacksquare \quad (\text{A.3})$$

Failure of Time-Reversal in Carli Arithmetic Mean (Formula Bias)

By contrast, the Carli arithmetic index fails this fundamental test due to Jensen's inequality:

$$I_{0,t}^{\text{Carli}} = \frac{1}{n} \sum_{i=1}^n \left(\frac{p_{i,t}}{p_{i,0}} \right), \quad I_{t,0}^{\text{Carli}} = \frac{1}{n} \sum_{i=1}^n \left(\frac{p_{i,0}}{p_{i,t}} \right) \quad (\text{A.4})$$

By the Cauchy-Schwarz and Arithmetic-Harmonic Mean inequalities:

$$I_{0,t}^{\text{Carli}} \times I_{t,0}^{\text{Carli}} \geq 1 \quad (\text{A.5})$$

with strict inequality holding whenever price ratios $p_{i,t}/p_{i,0}$ are not strictly identical across all items i , causing systematic upward compounding index drift.

Appendix D: Machine Learning Inflation Forecasting Feature Dictionary

Feature Column	Type	Economic / Econometric Concept	Mathematical Formulation
dod_pct	Float	Base daily percentage return	$r_t = \frac{\text{CPI}_t - \text{CPI}_{t-1}}{\text{CPI}_{t-1}} \times 100\%$
lag_{k}_dod ($k \in \{1, 2, 3, 7, 14, 30\}$)	Float	Autoregressive return memory	$L_k(t) = r_{t-k}$ (Strictly shifted by 1)
ma_7, ma_14, ma_30	Float	Trailing moving average trend	$\text{MA}_k(t) = \frac{1}{k} \sum_{i=1}^k \text{CPI}_{t-i}$
momentum_7_30	Float	Short-to-medium trend velocity	$\frac{\text{MA}_7(t) - \text{MA}_{30}(t)}{\text{MA}_{30}(t)} \times 100\%$
volatility_7d, 14d	Float	High-frequency price turbulence	$\sigma_k(t) = \sqrt{\frac{1}{k-1} \sum_{i=1}^k (r_{t-i} - \bar{r}_k)^2}$
food_momentum_7d	Float	Division 01 leading supply signal	$\frac{I_{\text{Food}, t-1} - I_{\text{Food}, t-8}}{I_{\text{Food}, t-8}} \times 100\%$
transport_momentum_7d	Float	Division 07 fuel/transit signal	$\frac{I_{\text{Trans}, t-1} - I_{\text{Trans}, t-8}}{I_{\text{Trans}, t-8}} \times 100\%$
month_sin, month_cos	Float	Annual cyclical Fourier terms	$\sin(\frac{2\pi \cdot m_t}{12}), \cos(\frac{2\pi \cdot m_t}{12})$
day_of_week, is_weekend	Int	Intra-week shopping cyclicity	$\text{DOW} \in [0, 6]$; Weekend dummy $\in \{0, 1\}$
is_holiday_window	Binary	Cambodian festival spending	Active window for KNY, Pchum Ben, Water Fest
target_{H}d ($H \in \{7, 14, 30\}$)	Float	Forward multi-horizon target	$y_t^{(H)} = \frac{\text{CPI}_{t+H} - \text{CPI}_t}{\text{CPI}_t} \times 100\%$

Table A.3: Production Feature Store Dictionary for Daily Machine Learning Inflation Forecasting